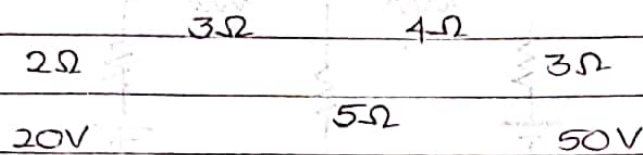


CHAPTER 1

REVIEW OF NETWORK ANALYSIS

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Q.1. Solve using mesh and nodal analysis to find the current in each loop and node voltage.

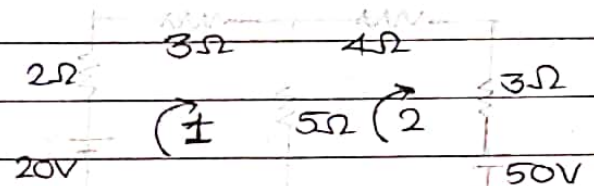
Sol: Applying KVL in loop (1):

$$20 - 2I_1 - 3I_1 - 5(I_1 - I_2) = 0$$

$$\Rightarrow 20 - 5I_1 - 5I_1 + 5I_2 = 0$$

$$\Rightarrow -10I_1 + 5I_2 = -20$$

$$\Rightarrow 2I_1 - I_2 = 4 \quad \dots (1)$$



Applying KVL in loop (2):

$$-5(I_2 - I_1) - 4I_2 - 3I_2 - 50 = 0$$

$$\Rightarrow -5I_2 + 5I_1 - 7I_2 - 50 = 0$$

$$\Rightarrow 5I_1 - 12I_2 = 50 \quad \dots (2)$$

Solving (1) and (2), $2I_1 - I_2 = 4$

$$5I_1 - 12I_2 = 50$$

$$I_1 = -0.105A \text{ and } I_2 = -4.21A.$$

Using nodal analysis:

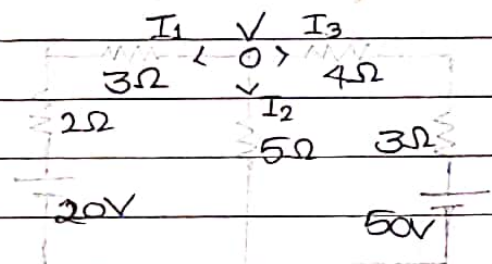
Applying KCL at node:

$$I_1 + I_2 + I_3 = 0$$

$$\Rightarrow \frac{V-20}{3+2} + \frac{V}{5} + \frac{V-50}{4+3} = 0$$

$$\Rightarrow \frac{V-20}{5} + \frac{V}{5} + \frac{V-50}{7} = 0$$

$$\Rightarrow \frac{2V-20}{5} + \frac{V-50}{7} = 0$$



$$\Rightarrow 14V - 140 + 5V - 250 = 0$$

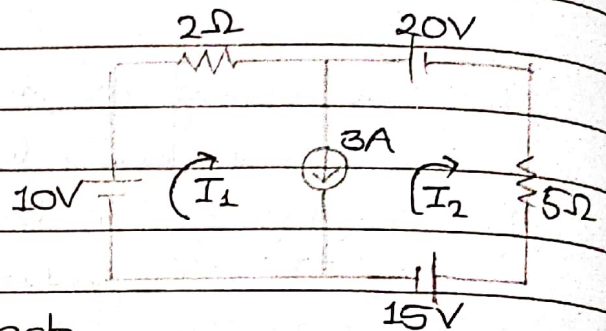
$$\Rightarrow 19V - 390 = 0$$

$$\therefore V = 20.53 V$$

Q.2. Find loop currents.

solⁿ: Here, $I_1 - I_2 = 3A$

$$\Rightarrow I_1 = 3 + I_2 \quad \dots (1)$$



Applying KVL in supermesh,

$$10 - 2I_1 - 20 - 5I_2 - 15 = 0$$

$$\Rightarrow -2I_1 - 5I_2 - 25 = 0$$

$$\Rightarrow 2I_1 + 5I_2 = -25$$

$$\Rightarrow 2(3 + I_2) + 5I_2 = -25 \quad \{ \because I_1 = 3 + I_2 \}$$

$$\Rightarrow 7I_2 = -25 - 6$$

$$\therefore I_2 = -4.43 A.$$

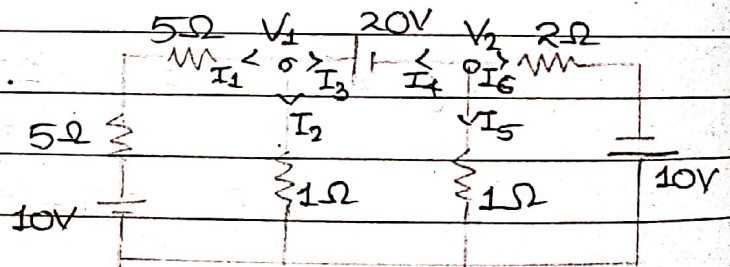
$$\text{And, from (1), } I_1 = 3 + (-4.43) = -1.43 A.$$

Q.3. Find node voltages.

solⁿ: Here,

$$V_1 - V_2 = 20$$

$$\therefore V_1 = 20 + V_2 \quad \dots (1)$$



Applying KCL at supernode, we get,

$$\frac{V_1 - 10}{5 + 5} + \frac{V_1}{1} + \frac{V_2 + 10}{2} + \frac{V_2}{1} = 0$$

$$\Rightarrow \frac{V_1 - 10}{10} + \frac{V_1}{1} + \frac{V_2 + 10}{2} + V_2 = 0$$

$$\Rightarrow \frac{V_1 + 10V_1 - 10}{10} + \frac{V_2 + 2V_2 + 10}{2} = 0$$

$$\Rightarrow \frac{11V_1 - 10}{10} + \frac{3V_2 + 10}{2} = 0$$

$$\Rightarrow 22V_1 - 20 + 30V_2 + 100 = 0$$

$$\Rightarrow 52(20 + V_2) + 30V_2 + 80 = 0$$

$$\Rightarrow 52V_2 + 520 = 0$$

$$\therefore V_2 = -10V$$

From eqⁿ (1), $V_1 = 20 - 10 = 10V$.

CHAPTER 2 CIRCUIT DIFFERENTIAL EQUATIONS

To study transient in electrical circuit, it is necessary to be familiar with the mathematical concept of differential equations and its solution techniques.

The order of differential equation represents higher order derivatives and equal to the number of energy storing elements. The differential operator is denoted by D i.e. $\frac{d}{dt} = D$, $\frac{d^2}{dt^2} = D^2$.

$$* V_L = L \frac{d}{dt} i(t)$$

$$* V_C = \frac{1}{C} \int i(t) dt$$

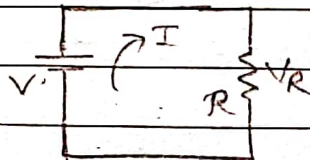
Operational Impedance -

1. Resistor -

$$V - V_R = 0$$

$$\Rightarrow V = IR$$

$$\Rightarrow IZ = IR \therefore Z = R$$

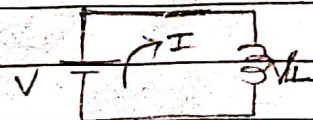


2. Inductor -

$$V - V_L = 0$$

$$\Rightarrow V - L \frac{dI}{dt} = 0 \Rightarrow V - LDI = 0$$

$$\Rightarrow IR = LDI \therefore Z = LD$$



$\therefore$ Operational impedance offered by inductor (Z) = LD .

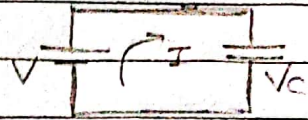
3. Capacitor -

$$V + V_c = 0$$

$$\Rightarrow V = -V_c \Rightarrow V_c = -\frac{1}{C} \int i(t) dt$$

$$\Rightarrow I = C \frac{dV_c(t)}{dt}$$

$$\Rightarrow I = CV_c D \Rightarrow I = \frac{V_c}{CD} \therefore I = \frac{V_c}{Z}$$



* Find the operational impedance of RLC series circuit.

Sol. Here, $Z_D = R + LD + \frac{1}{CD}$



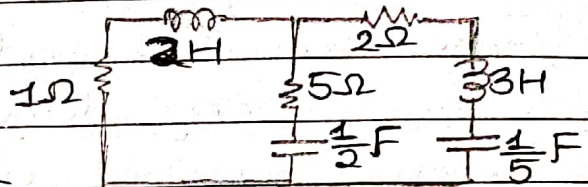
$$\therefore Z_D = \frac{RCD + LCD^2 + 1}{CD}$$

1. Find Z_0 .

Sol. Here, $Z_1 = 1 + 2D$

$$Z_2 = 5 + \frac{2}{D} = \frac{5D + 2}{D}$$

$$Z_3 = 2 + 3D + \frac{5}{D} = \frac{2D + 3D^2 + 5}{D}$$



$$\therefore Z_D = (Z_1 || Z_2) + Z_3$$

$$= \frac{(1+2D) \times \left(\frac{5D+2}{D}\right)}{(1+2D) + \left(5 + \frac{2}{D}\right)} + \frac{2D+3D^2+5}{D}$$

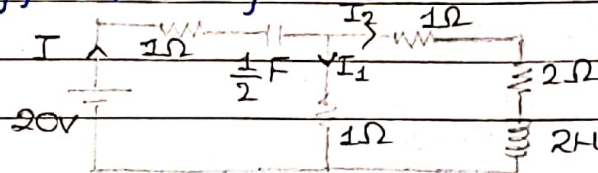
$$= \frac{5D + \frac{4D}{D} + \frac{10D^2+2}{D}}{\frac{D+2D^2+5D+2}{D}} + \frac{2D+3D^2+5}{D}$$

$$= \frac{10D^2+9D+2}{2D^2+6D+2} + \frac{2D+3D^2+5}{D}$$

$$= \frac{10D^3+9D^2+2D+4D^3+6D^4+10D^2+12D^2+18D^3+30D}{2D^3+6D^2+2D}$$

$$\therefore Z_D = \frac{6D^4 + 32D^3 + 37D^2 + 36D + 10}{2D^3 + 6D^2 + 2D}$$

2. Determine current I, I_1, I_2 and voltage V_2 in terms of differential operator.

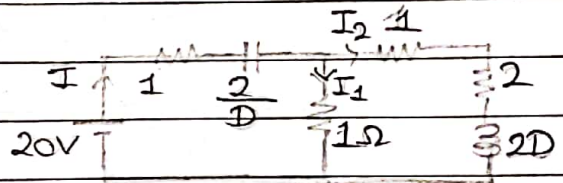


Sol: Replacing RLC using D-operator:

$$\text{Here, } Z_1 = 1 + \frac{2}{D}$$

$$Z_2 = 1$$

$$Z_3 = (1+2) + 2D = 3 + 2D$$



$$\therefore Z_D = Z_2 \parallel Z_3 + Z_1$$

$$= \frac{(3+2D) \times 1}{3+2D+1} + \left(\frac{D+2}{D} \right)$$

$$= \frac{3+2D}{2D+4} + \frac{D+2}{D}$$

$$= \frac{3D + 2D^2 + 2D^2 + 4D + 4D + 8}{2D^2 + 4D}$$

$$\therefore Z_D = \frac{4D^2 + 11D + 8}{2D^2 + 4D}$$

$$\text{Now, } I = \frac{V}{Z_D} = \frac{20(2D^2 + 4D)}{4D^2 + 11D + 8} = \frac{40D^2 + 80D}{4D^2 + 11D + 8}$$

$$I_1 = I \times \frac{Z_3}{Z_2 + Z_3} = \left(\frac{40D^2 + 80D}{4D^2 + 11D + 8} \right) \times \left(\frac{3+2D}{1+3+2D} \right)$$

$$\therefore I_1 = \frac{120D^2 + 80D^3 + 240D + 160D^2}{(4D^2 + 11D + 8)(2D + 4)} = \frac{280D^2 + 80D^3 + 240D}{(4D^2 + 11D + 8)(2D + 4)}$$

$$\text{And, } I_2 = \frac{I \times Z_2}{Z_2 + Z_3}$$

$$= \left(\frac{40D^2 + 11D + 8}{2D^2 + 4D} \right) \times \left(\frac{1}{4 + 2D} \right)$$

$$= \frac{40D^2 + 11D + 8}{8D^2 + 4D^3 + 16D + 8D^2}$$

$$\therefore I_2 = \frac{40D^2 + 11D + 8}{4D^3 + 16D^2 + 16D}$$

$$\text{Now, } V_2 = I_2 Z_2 = \left(\frac{40D^2 + 11D + 8}{4D^3 + 16D^2 + 16D} \right) \times 1$$

$$\therefore V_2 = \frac{40D^2 + 11D + 8}{4D^3 + 16D^2 + 16D}$$

Solution of the differential equation -

Let the differential equation be in the form

$$y(t) = G(D) \times f(t) \quad \dots (1)$$

where, $y(t) \rightarrow$ response

$G(D) \rightarrow$ network function with D -operator

$f(t) \rightarrow$ forcing function/excitation function

In classical method, eqⁿ (1) has steady state solution and transient solution.

* Steady state solution:

1) If $f(t) = (\text{constant}) k$

$$y_{ss}(t) = G(D) |_{D=0}$$

2) If $f(t) = A e^{st}$

$$y_{ss}(t) = G(D) |_{D=s}$$

3) If $f(t) = Ae^{jst}$

$$y_{ss}(t) = A[\cos st + j \sin st]$$

$$= \operatorname{Re}[A \cos st] + \operatorname{Im}[A j \sin st]$$

4) If $f(t) = \sin \omega t$

$$y_{ss}(t) = G(D) \big|_{D=j\omega} \operatorname{Im}[Ae^{j\omega t}]$$

5) If $f(t) = \cos \omega t$

$$y_{ss}(t) = G(D) \big|_{D=j\omega} \operatorname{Re}[Ae^{j\omega t}]$$

* Transient solution:

$$G(D) = \frac{\text{Numerator}(D)}{\text{Denominator}(D)}$$

Characteristic equation is equated with zero

$$\text{i.e. Denominator}(D) = 0.$$

$$\text{Roots, } D = s_1, s_2$$

Case 1: If s_1 and s_2 are real and unequal.

$$y_{Ts}(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

Case 2: If s_1 and s_2 are real and equal.

$$y_{Ts}(t) = k_1 e^{s_1 t} + t k_2 e^{s_2 t}$$

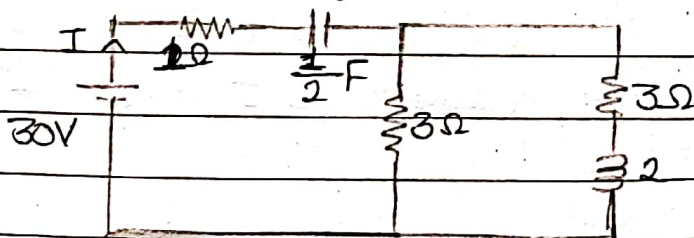
Case 3: If s_1 and s_2 are imaginary.

$$D = s_1 \pm j s_2 [\alpha \pm j \beta]$$

$$y_{Ts}(t) = \alpha [k_1 \cos \beta t + j k_2 \sin \beta t]$$

$$\therefore y(t) = y_{Ts} + y_{ss}$$

Q.1. Obtain the complete solution for the given circuit.

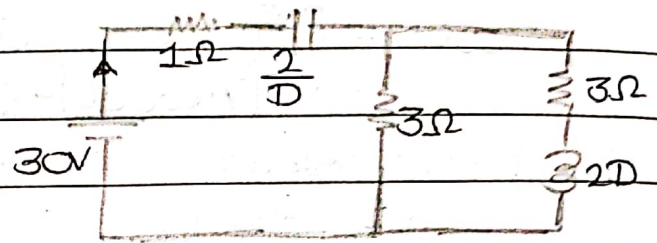


Ans: Replacing RLC using D-operator:

$$\text{Here, } Z_1 = 1 + \frac{2}{D} = \frac{D+2}{D}$$

$$Z_2 = 3$$

$$Z_3 = 3 + 2D$$



$$\text{Then, } Z_D = Z_2 \parallel Z_3 + Z_1$$

$$= \left\{ \frac{3 \times (3 + 2D)}{3 + 3 + 2D} \right\} + \left(\frac{D+2}{D} \right)$$

$$= \frac{9 + 6D}{2D + 6} + \frac{D+2}{D}$$

$$\Rightarrow Z_D = \frac{9D + 6D^2 + 2D^2 + 4D + 6D + 12}{2D^2 + 6D}$$

$$\therefore Z_D = \frac{8D^2 + 19D + 12}{2D^2 + 6D}$$

$$\text{Now, } I = \frac{V}{Z_D} = \frac{30(2D^2 + 6D)}{8D^2 + 19D + 12}$$

$$\text{For S.S.S: } f(t) = 30$$

$$I_{ss}(t) = \frac{(2D^2 + 6D)}{8D^2 + 19D + 12} \times 30 \quad | \quad D=0$$

$$\therefore I_{ss}(t) = 0 \quad \dots (1)$$

$$\text{For T.S: } D(6) = 0$$

$$16D^2 + 8D^2 + 19D + 12 = 0$$

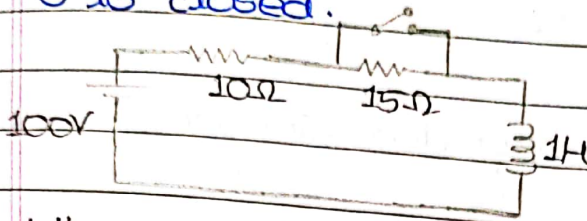
$$\therefore D = -0.209 \pm 1.79j \quad -1.1875 \pm 0.299j$$

$$\therefore I_{ss}(t) = \{ k_1 e^{-0.209t} + k_2 e^{-1.79t} \} \quad \dots (2)$$

$$= e^{-1.1875t} [A \cos 0.299t + B \sin 0.299t]$$

$$\text{Thus, the required solution is } I(t) = k_1 e^{-0.209t} + k_2 e^{-1.79t}$$

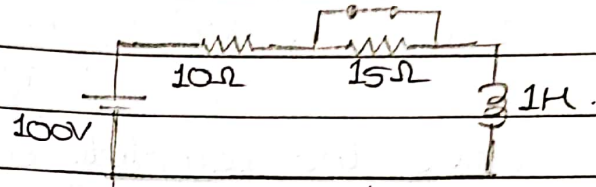
2. Obtain the complete solution for current when switch 'S' is closed.



solⁿ When switch is closed, the circuit becomes,
Here, $Z_D = 10 + D$

Now,

$$\text{Current (I)} = \frac{V}{Z_D} \\ = \frac{100}{10 + D}$$



For S.S solⁿ:

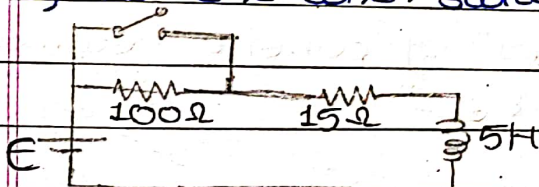
$$\hat{I}_{ss}(t) = G(D) \mid_{D=0} \\ \Rightarrow \hat{I}_{ss}(t) = 10 \quad \text{--- (1)}$$

For transient solution:

$$I_{ss}(t) = G(t) \quad 10 + D = 0 \quad \therefore D = -10 \\ \therefore I_{ts}(t) = k e^{-10t} \quad \text{--- (2)}$$

Thus, the complete solution for current is $I(t) = 10 + k e^{-10t}$

3. If $E = 100V$ and $E = e^{-6t}$, find the complete solⁿ of current when switch is closed using D-operator.

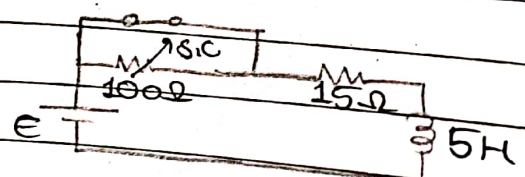


solⁿ When the switch is closed, the ckt becomes,

Replacing the ckt with

D-operator,

$$Z_D = 15 + 5D$$



i) When $E = 100V$, $I = \frac{100}{15 + 5D}$

For S.S solⁿ: $i_{ss}(t) = G(D) \big|_{D=0}$
 $\Rightarrow i_{ss}(t) = 6.66 \dots (1)$

For T.S 8- ~~is~~ Equating denominator to zero,
 $15 + 5D = 0$
 $\therefore D = -3$

$\therefore i_{TS}(t) = k e^{-3t} \dots (2)$

Thus, the complete solⁿ of current becomes,
 $i(t) = 6.66 + k e^{-3t}$.

ii) When $E = e^{-6t}$, $I = \frac{e^{-6t}}{15 + 5D}$

For S.S solⁿ: $i_{ss}(t) = G(D) \big|_{D=5}$
 $\Rightarrow i_{ss}(t) = \frac{e^{-6t}}{15 + (5 \times -6)}$

$\therefore i_{ss}(t) = \frac{e^{-6t}}{-15} = -0.067 e^{-6t} \dots (3)$

For transient solⁿ: Equating denominator to zero,
 $15 + 5D = 0 \therefore D = -3$

$\therefore i_{TS}(t) = k e^{-3t} \dots (4)$

Thus, the complete solⁿ of current becomes,
 $i(t) = 0.667 e^{-6t} + k e^{-3t}$.

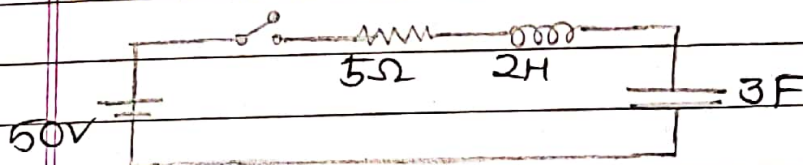
Initial conditions in the circuit - We need initial conditions for determining the arbitrary constants in the general solution of differential equation. The number of initial conditions required is equal to the order of differential eqⁿ for determining the solution.

Following steps are used for initial condition -

- 1) Replace all inductors with open ckt at $t=0^+$.
- 2) Replace all capacitors with short ckt at $t=0^+$.
- 3) Resistors are connected in the ckt without any change.

Note that the current in the circuit does not change instantly if inductor is present in the ckt and voltage doesn't change instantly if the capacitor is present in the circuit.

Q.1. Find the complete solution of the current when switch 'S' is closed at $t=0$.



solⁿ Applying KVL in the given circuit when switch is closed, we have,

$$50 - 5i(t) + 2 \frac{d i(t)}{dt} + \frac{1}{3} \int i(t) dt = 50 \quad \dots (1)$$

$$\text{Differentiating (1), } 5 \frac{d i(t)}{dt} + 2 \frac{d^2 i(t)}{dt^2} + \frac{1}{3} i(t) = 0 \quad \dots (2)$$

Let $\frac{d^2 i(t)}{dt^2} = p^2$, $\frac{di(t)}{dt} = p$.

Then, $2p^2 + 5p + \frac{1}{3} = 0$

$\Rightarrow 6p^2 + 15p + 1 = 0$

$\Rightarrow p = \frac{-15 \pm \sqrt{225 - 24}}{12}$

$\therefore p = -0.068, -2.43$

$\therefore i(t) = k_1 e^{-0.068t} + k_2 e^{-2.43t} \dots (2)$

Applying initial condition, $i(0) = 0$

$i(0^+) = 0$

$\Rightarrow k_1 + k_2 = 0 \dots (3)$

From (1), $2 \frac{di(0)}{dt} = 50$

$\therefore \frac{di(0)}{dt} = 25 \text{ A/s.}$

Differentiating (2), $\frac{di(t)}{dt} = -0.068k_1 e^{-0.068t} - 2.43k_2 e^{-2.43t}$

At $t=0$, $\frac{di(0)}{dt} = -0.068k_1 - 2.43k_2$

$\Rightarrow -0.068k_1 - 2.43k_2 = 25 \dots (4)$

Solving (3) and (4), $k_1 = 10.58$ & $k_2 = -10.58$

$\therefore i(t) = 10.58 e^{-0.068t} - 10.58 e^{-2.43t}$

Equivalent ckt of RLC at initial condition -

Elements

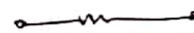
Equivalent ckt
at $t=0^+$

Equivalent ckt
at $t=\infty$

S.N.

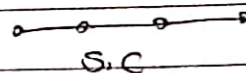
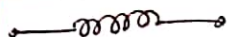
1.

Resistor



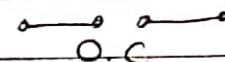
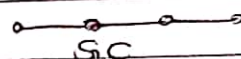
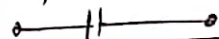
2.

Inductor



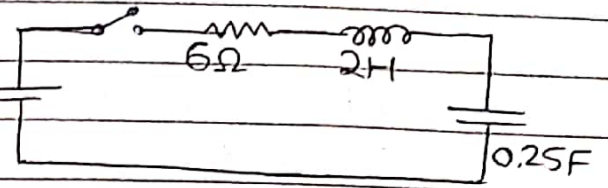
3.

Capacitor



Q.2. For the given ckt, determine $i(0^+)$, $\frac{d}{dt} i(0^+)$, $\frac{d^2}{dt^2} i(0^+)$ and $i(t)$.

solⁿ Applying KVL in the given ckt when switch is closed at $t=0^+$,



$$6i(t) + 2 \frac{d}{dt} i(t) + \frac{1}{0.25} \int i(t) dt = 2 \quad \dots (1)$$

Diff. (1) w.r.t t , we get,

$$\Rightarrow 6 \frac{d}{dt} i(t) + 2 \frac{d^2}{dt^2} i(t) + 4i(t) = 0 \quad \dots (2)$$

$$\text{Let } \frac{d^2}{dt^2} i(t) = p^2, \quad \frac{d}{dt} i(t) = p.$$

$$2p^2 + 6p + 4 = 0$$

$$\Rightarrow p^2 + 3p + 2 = 0 \Rightarrow (p+1)(p+2) = 0$$

$$\Rightarrow p = -1, -2.$$

$\therefore$ The general solution of the differential eqⁿ is

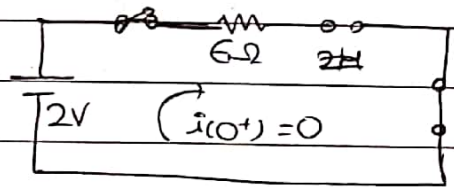
$$i(t) = k_1 e^{-t} + k_2 e^{-2t} \quad \dots (3)$$

Applying initial condition,

$$i(0^+) = 0$$

$$\text{From (1), } 2 \frac{d}{dt} i(0^+) = 2$$

$$\therefore \frac{d}{dt} i(0^+) = 1 \text{ A/s.}$$



$$\text{From (2), } 6 \times 1 + 2 \frac{d^2}{dt^2} i(0^+) + 0 = 0$$

$$\therefore \frac{d^2}{dt^2} i(0^+) = -3 \text{ A/s}^2$$

$$\text{At } t=0^+, \quad i(0^+) = 0$$

$$\Rightarrow k_1 + k_2 = 0 \quad \dots (4)$$

$$\text{Again diff. (3), } \frac{d}{dt} i(t) = -k_1 e^{-t} - 2k_2 e^{-2t}$$

$$\therefore \frac{d}{dt} i(0^+) = -k_1 - 2k_2 \Rightarrow -k_1 + 2k_2 = -1 \quad \dots (5)$$

Solving (4) and (5), $k_1 + k_2 = 0$

$$-k_1 + 2k_2 = -1$$

$$-0k_2 = 1$$

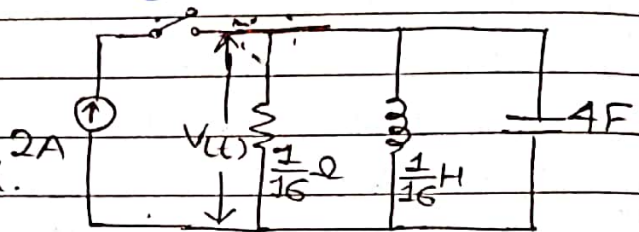
$\therefore k_2 = -1$ and $k_1 = 1$.

Thus, the particular solution becomes,

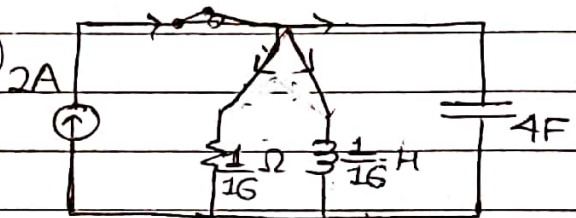
$$i(t) = e^{-t} - e^{-2t}$$

3. Consider a RLC circuit as shown below and determine $V(0^+)$, $\frac{d}{dt} V(0^+)$, $\frac{d^2}{dt^2} V(0^+)$ and $V(t)$.

Q2) Applying KCL in the given ckt when the switch is closed.



$$\frac{1}{16} + \frac{1}{16} \int V(t) dt + 4 \frac{d}{dt} V(t) = 2 \quad \text{--- (1)}$$



Differentiating (1) w.r.t t ,

$$\frac{1}{16} + \frac{1}{16} \frac{d}{dt} V(t) + 16 V(t) + 4 \frac{d^2}{dt^2} V(t) = 0 \quad \text{--- (2)}$$

Let $\frac{d^2}{dt^2} V(t) = p^2$ and $\frac{d}{dt} V(t) = p$, then

$$4p^2 + 16p + 16 = 0$$

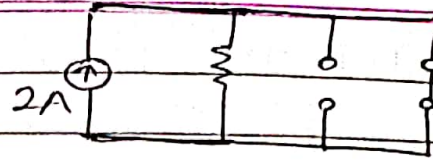
$$\Rightarrow p^2 + 4p + 4 = 0$$

$$\Rightarrow p^2 + 2p + 2p + 4 = 0 \Rightarrow (p+2)(p+2) = 0 \therefore p = -2, -2$$

$\therefore$ The general solⁿ of differential equation is

$$V(t) = k_1 e^{-2t} + k_2 t e^{-2t} \quad \text{--- (3)}$$

Applying initial condition,
 $V(0^+) = 0$



From (1), $4 \frac{dV(0^+)}{dt} = 2$

$\therefore \frac{dV(0^+)}{dt} = \frac{1}{2} \text{ V/s}$

From (2), $16 \times \frac{1}{2} + 16 \times 0 + 4 \frac{d^2 V(0^+)}{dt^2} = 0$

$\Rightarrow 8 + 4 \frac{d^2 V(0^+)}{dt^2} = 0$

$\therefore \frac{d^2 V(0^+)}{dt^2} = -2 \text{ V/s}^2$

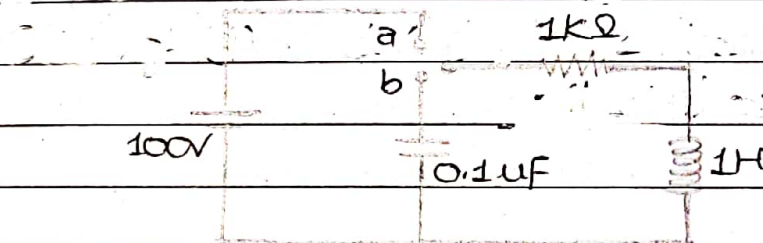
At $t=0^+$, $V(0^+) = 0$ And, $V'(0^+) = \frac{1}{2}$

$\Rightarrow k_1 = 0 \quad \Rightarrow k_2 = \frac{1}{2}$

$\therefore$ The particular solution becomes,

$V(t) = \frac{1}{2} e^{-2t}$

4. In the given ckt, switch S is changed from position a to b at time $t=0$, find the values of current i , $\frac{d i(0^+)}{dt}$, $\frac{d^2 i(0^+)}{dt^2}$ at $t=0^+$.



$\therefore$ At $t=0$, switch is connected to point b.

Applying KVL in the ckt,

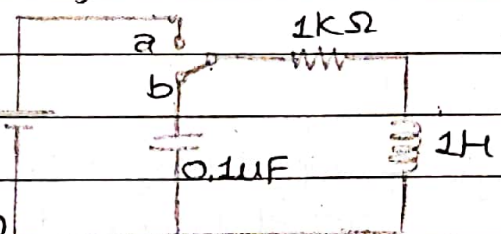
$\frac{1}{C} \int i(t) dt + i(t)R + L \frac{d i(t)}{dt} = 0$

$\Rightarrow \frac{1}{0.1 \times 10^{-6}} \int i(t) dt + i(t) \times 10^3 + \frac{d i(t)}{dt} = 0$

$\Rightarrow 10^7 \int i(t) dt + 10^3 i(t) + \frac{d i(t)}{dt} = 0 \quad \dots (1)$

Differentiating (1) w.r.t t ,

$10^7 i(t) + 10^3 i'(t) + i''(t) = 0 \quad \dots (2)$



Let $i'(u) = p$ and $i''(u) = p^2$. Then

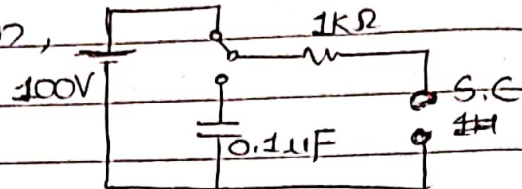
$$p^2 + 10^3 p + 10^7 = 0$$

$$\therefore p = -500 \pm 3122.49j$$

$$\therefore i(u) = e^{-500t} (K_1 \cos 3122.49t + K_2 \sin 3122.49t)$$

Applying initial condition,

$$i(0) = \frac{100}{1000} = 0.1 \text{ A}$$



$$i(0^-) = i(0) = i(0^+) = 0.1 \text{ A}$$

$$\text{From (1), } 10^3 \times 0.1 + \frac{di(0)}{dt} = 0$$

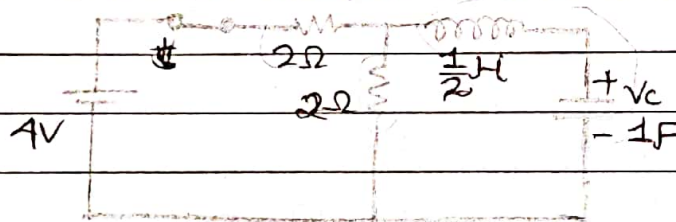
$$\therefore \frac{di(0^+)}{dt} = -100 \text{ A/s}$$

$$\text{From (2), } 10^7 \times 0.1 + 10^3 \times (-100) + \frac{d^2 i(0^+)}{dt^2} = 0$$

$$\Rightarrow \frac{d^2 i(0^+)}{dt^2} = -9 \times 10^5 \text{ A/s}^2$$

At $t=0^+$,

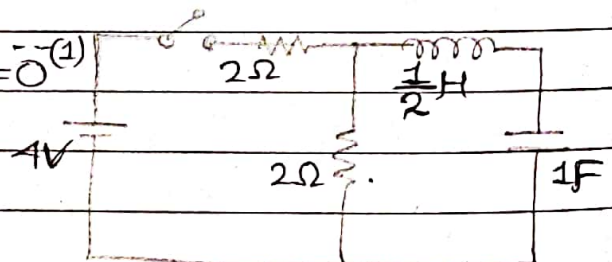
5. The circuit given below is a steady state with switch closed. The switch is opened at $t=0$, determine the current across inductor.



sol Applying KVL in the circuit when switch is opened at $t=0$, we have,

$$V_L(t) + 2i(t) + \frac{1}{2} \frac{di(t)}{dt} + \int i(t) dt = 0 \quad (1)$$

Differentiating (1) w.r.t t ,



$$-2(0-1) = V_c$$

$$\therefore V_c = 2V.$$

$$2i'(t) + \frac{1}{2}i''(t) + i(t) = 0 \quad \dots (2)$$

Let $i'(t) = p$ and $i''(t) = p^2$. Then,

$$2p + \frac{1}{2}p^2 + 1 = 0$$

$$\Rightarrow p^2 + 4p + 2 = 0 \quad \therefore p = -0.58, -3.41$$

$\therefore$ The general solution of differential equation is

$$i(t) = k_1 e^{-0.58t} + k_2 e^{-3.41t} \quad \dots (3)$$

Applying initial condition,

$$i(0^-) = \frac{4}{2+2} = 1A.$$

$$\text{And, } i(0^-) = 0A$$

$\therefore$ Inductor is present in the ckt. $i(0^-) = i(0) = i(0^+) = 1A.$

$$\text{At } t=0^+, i_L(0^+) = 0A/s \quad 0A/s$$

$$\text{From (1), } i_L'(0^+) = 0A/s \cdot -2 \times 2 = -4A/s$$

$$\text{At } t=0^+, \text{ from eqn (3), } 0 = k_1 + k_2$$

$$\therefore k_1 = -k_2 \quad \dots (4)$$

$$\text{From Differentiating (3), } i'(t) = -0.58k_1 e^{-0.58t} - 3.41k_2 e^{-3.41t}$$

$$\Rightarrow i'(0^+) = -0.58k_1 - 3.41k_2$$

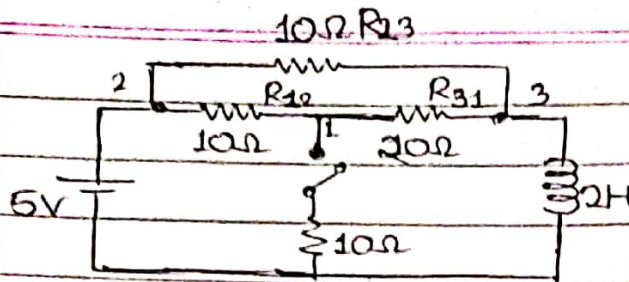
$$\Rightarrow -0.58k_1 - 3.41k_2 = -4$$

$$\Rightarrow +0.58k_2 - 3.41k_2 = -4$$

$$\therefore k_2 = -1.41 \quad \text{and } k_1 = 1.41$$

$$\therefore i(t) = 1.41 e^{-0.58t} - 1.41 e^{-3.41t}$$

6. In the ckt, the switch is open and the circuit reaches a steady state. At $t=0$, the switch is closed, find the current in the inductor for $t>0$.

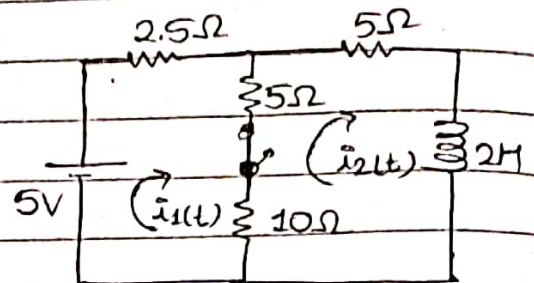


8/1/20
Converting delta to star,

$$R_1 = \frac{10 \times 20}{10 + 20 + 10} = 5\Omega$$

$$R_2 = \frac{10 \times 10}{40} = 2.5\Omega$$

$$R_3 = \frac{10 \times 20}{40} = 5\Omega$$



When switch is closed, applying KVL in the loop (1),

$$5 - 2.5i_1(t) - 15i_1(t) + 15i_2(t) = 0$$

$$\Rightarrow 17.5i_1(t) - 15i_2(t) = 5 \quad \dots (1)$$

Applying KVL in the loop (2),

$$-15i_2(t) + 15i_1(t) - 5i_2(t) - 2 \frac{d}{dt} i_2(t) = 0$$

$$\Rightarrow 15i_1(t) - 20i_2(t) - 2i_2'(t) = 0 \quad \dots (2)$$

$$\Rightarrow \text{From (1), } i_1(t) = \frac{5\{1 + 3i_2(t)\}}{17.5}$$

Substituting this value in (2),

$$\frac{15 \times 5}{17.5} (1 + 3i_2(t)) - 20i_2(t) - 2i_2'(t) = 0$$

$$\Rightarrow 2i_2'(t) + 20i_2(t) = \frac{30}{7} + \frac{90}{7}i_2(t)$$

$$\Rightarrow 2i_2'(t) + \frac{50}{7}i_2(t) = \frac{30}{7}$$

$$\Rightarrow i_2'(t) + \frac{25}{7}i_2(t) = \frac{15}{7} \quad \dots (3)$$

The general solution of (3) is

$$i_2(t) = \frac{15/7}{25/7} + k e^{-25/7 t}$$

$$\therefore i_2(t) = \frac{3}{5} + k e^{-25/7 t} \quad \dots (4)$$

Applying initial condition,

$$i_2(0) = \frac{5}{2.5+5} = 0.67 \text{ A.}$$

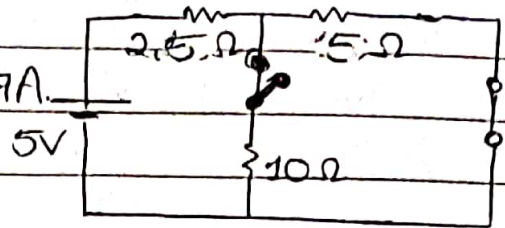
$$\text{At } t=0^+, i_2(0^-) = i_2(0) = i_2(0^+) = 0.67 \text{ A.}$$

From (4), at

$$0.67 = \frac{3}{5} + k$$

$$\therefore k = 0.07.$$

$$\therefore i_2(t) = \frac{3}{5} + 0.07 k e^{-25/7 t} \text{ A.}$$



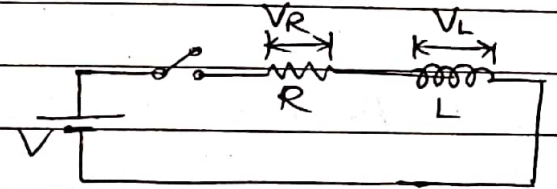
CHAPTER 3 CIRCUIT DYNAMICS

Response of RL series circuit -

Let us consider RL series circuit with the supplied voltage V . To determine the response of RL series circuit i.e current of the ckt, we apply KVL in the loop when switch is closed.

Applying KVL in the ckt,

$$V - V_R - V_L = 0$$



$$\Rightarrow V - i(t)R - L \frac{d}{dt} i(t) = 0$$

$$\Rightarrow \frac{d}{dt} i(t) + i(t) \frac{R}{L} = \frac{V}{L} \quad \dots (1)$$

Comparing with the standard form,

$\frac{d}{dt} y(t) + P y(t) = Q$, solution of this eqⁿ is given by,

$$y(t) = \frac{Q}{P} + k e^{-Pt} \quad \dots (2)$$

$$\therefore i(t) = \frac{V/L}{R/L} + k e^{-Pt} = \frac{V}{R} + k e^{-Pt} \quad \dots (2)$$

Applying initial condition, at $t=0^+$,

$$i(0) = \frac{V}{R} + k$$

$$\therefore k = -\frac{V}{R}$$

$$\therefore \text{Eq (2) becomes } i(t) = \frac{V}{R} - \frac{V}{R} e^{-R/L t}$$

$$\therefore i(t) = \frac{V}{R} (1 - e^{-R/L t}) \quad \dots (a)$$

$$\text{Now, } V_R = i(t) R = \frac{V}{R} (1 - e^{-R/L t}) R$$

$$\therefore V_R = V (1 - e^{-R/L t}) \quad \dots (b)$$

$$\text{And, } V_L = L \frac{di(t)}{dt} = -L \cdot \frac{V}{R} \cdot \left(-\frac{R}{L}\right) e^{-R/L t}$$

$$\therefore V_L = V e^{-R/L t} \quad \dots (c)$$

$$\text{OR, } V_L = V - V_R = V - V + V e^{-R/L t} = V e^{-R/L t}$$

Case I: Time constant: $t = \tau = \frac{L}{R}$.

$$i(\tau) = 0.632 \frac{V}{R} = 63.2\% \text{ of } \frac{V}{R}$$

$$V_R(\tau) = V (1 - e^{-1}) = 0.632 V = 63.2\% \text{ of } V$$

$$V_L(\tau) = V e^{-1} = 0.367 V = 36.7\% \text{ of } V$$

Case II: At $t=0$, $i(0) = 0$

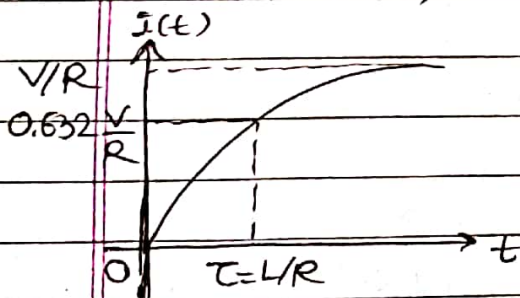
$$V_R(0) = 0$$

$$V_L(0) = V$$

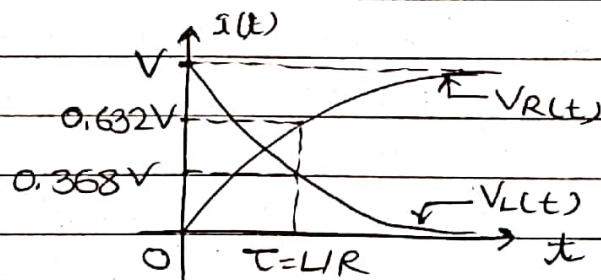
Case III: At $t=\infty$, $i(\infty) = \frac{V}{R}$

$$V_R(\infty) = V$$

$$V_L(\infty) = 0.$$



for $i(t)$

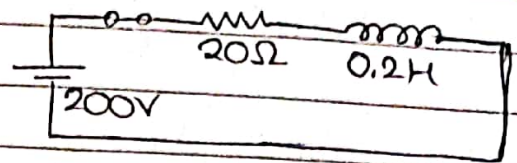


for $V_R(t)$ and $V_L(t)$

1. A DC voltage of 200V is applied to a series RL ckt with resistance of 20Ω and inductor of $0.2H$. Determine the voltage across inductor at instant of switching and 0.02 sec later.

Sol: Applying KVL in the ckt,

$$200 - 20i(t) + 0.2 i'(t) = 0$$



$$\Rightarrow i'(t) + 100 i(t) = 1000 \quad \dots (1)$$

The solⁿ of differential eqⁿ (1) is given by,

$$i(t) = \frac{1000}{100} + k e^{-100t} = 10 + k e^{-100t}$$

Applying initial condition,

$$i(0) = 0$$

$$\therefore 0 = 10 + k \Rightarrow k = -10$$

$$\therefore i(t) = 10(1 - e^{-100t})$$

Now, Voltage across inductor $V_L(t) = 200 - 200 + 200 e^{-100t}$

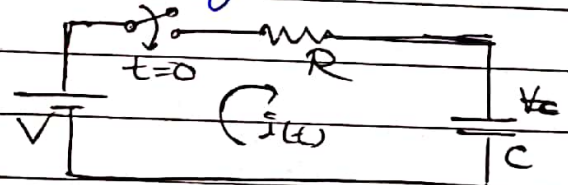
$$\therefore V_L(t) = 200 e^{-100t}$$

$$V_L(0.02) = 200 e^{-100 \times 0.02} = 27.07 \text{ V}$$

Response of series RC-circuit:

Applying KVL in the ckt,

$$V - i(t)R - \frac{1}{C} \int i(t) dt = 0$$



$$\Rightarrow i(t)R + \frac{1}{C} \int i(t) dt = V_{C(0+)} = V \quad \dots (1)$$

$\Rightarrow$ Differentiating w.r.t t , we get,

$$i'(t)R + \frac{1}{C} i(t) = 0 \quad \dots (2)$$

$$\Rightarrow i'(t) + \frac{1}{RC} i(t) = 0$$

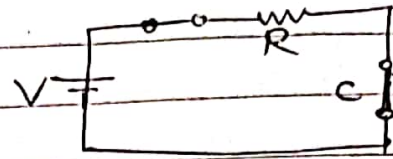
The general solution of eqⁿ (2) is given by,

$$i(t) = \frac{V}{R} k e^{-\frac{t}{RC}} \quad \dots (3)$$

Applying initial condition, at $t=0^+$,

$$i(0^+) = k \Rightarrow k = \frac{V}{R}$$

$$\therefore i(t) = \frac{V}{R} e^{-\frac{t}{RC}} \quad \dots (a)$$



$$\text{Now, } V_C(t) = V - V_R = V - i(t)R$$

$$= V - \frac{V}{R} \cdot R e^{-t/RC}$$

$$\therefore V_C(t) = V(1 - e^{-t/RC}) \quad \dots (b)$$

$$\text{And, } V_R(t) = V e^{-t/RC} \quad \dots (c)$$

Case I: At $t=0$, $i(0) = \frac{V}{R}$

$$V_R(0) = V$$

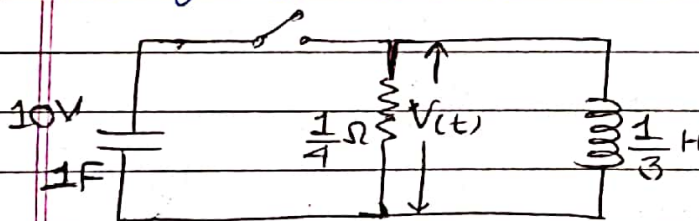
$$V_C(0) = 0$$

Case II: At $t=\infty$, $i(\infty) = 0$

$$V_R(\infty) = 0$$

$$V_C(\infty) = V$$

Q. 1. In the given circuit, the capacitor has initial charge of $10V$ and at same instant, current through inductor is zero, find the expression for voltage across inductor using classical method.



solⁿ When switch is closed, applying KVL to the ckt,

$$10 - \int i(t) dt - V(t) = 0 \quad \dots (1)$$

Differentiating (1), $i(t) = -\frac{dV(t)}{dt} \quad \dots (2)$

Applying KCL at node, $i(t) = i_R(t) + i_L(t)$

$$\Rightarrow +C \left(\frac{dV(t)}{dt} \right) + \frac{V(t)}{R} + \frac{1}{L} \int V(t) dt = 0$$

$$\Rightarrow +V'(t) + 4V(t) + 3 \int V(t) dt = 0 \Rightarrow V'(t) + 4V(t) + 3 \int V(t) dt = 0$$

Differentiating, $V''(t) + 4V'(t) + 3V(t) = 0$

Let $V''(t) = p^2$ and $V'(t) = p$. Then,

$$p^2 + 4p + 3 = 0$$

$$\therefore p = -1, -3$$

The general solution of above differential eq is

$$V(t) = k_1 e^{-t} + k_2 e^{-3t} \dots (3)$$

$$\text{At } t=0^+, V(0^+) = k_1 + k_2$$

$$\therefore k_1 + k_2 = 10 \dots (4)$$

$$\text{Diff. (3), } V'(t) = -k_1 e^{-t} - 3k_2 e^{-3t}$$

$$\text{At } t=0^+, V'(0^+) = -k_1 - 3k_2$$

$$\Rightarrow 0 = k_1 + 3k_2 \dots (5) \quad (\because V' \text{ is initially 0})$$

Solving eqn (4) and (5),

$$k_1 + k_2 = 10$$

$$-k_1 + 3k_2 = 0$$

$$-2k_2 = 10 \therefore k_2 = -5 \text{ and } k_1 = 15$$

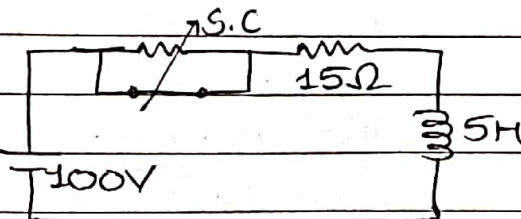
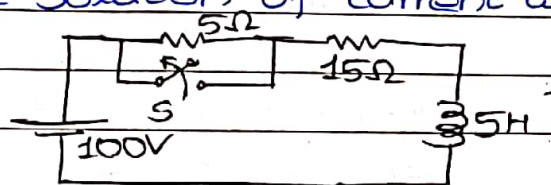
$$\therefore \text{From eqn (3), } V(t) = 15 e^{-t} - 5 e^{-3t}$$

2. Determine the complete solution of current when switch 'S' is closed.

sol: When switch is closed, applying KVL to the ckt,

$$100 - 15i(t) - 5i'(t) = 0$$

$$\Rightarrow 5i'(t) + 3i(t) = 20 \dots (1)$$



The general solution of eqⁿ (1) is

$$i(t) = \frac{20}{3} + k e^{-\frac{20}{3}t} \quad \dots (2)$$

Applying initial condition,

$$i(0^-) = \frac{100}{5+15} = \frac{100}{20} = 5 \text{ A.}$$

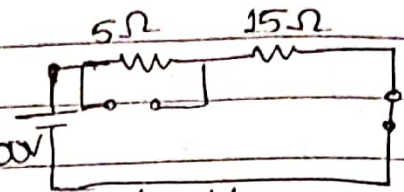
Since inductor is present in 100V the ckt, current doesn't change instantly.

$$\text{So, } i(0^-) = i(0) = i(0^+) = 5 \text{ A.}$$

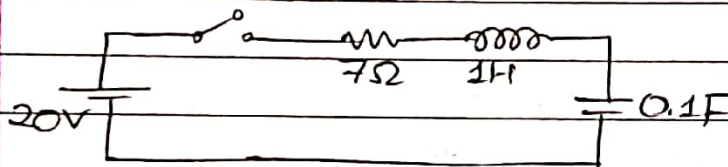
$$\text{From (2), } 5 = \frac{20}{3} + k \quad \therefore k = -\frac{5}{3}$$

$\therefore$ The required complete solⁿ of current is,

$$i(t) = \frac{20}{3} - \frac{5}{3} e^{-3t}$$



3. Determine the expression for current when the switch is closed at $t=0$. Assuming initial voltage across capacitor to be zero and initial current across inductor is zero.



Applying KVL in the ckt when switch is closed,

$$20 - 7i(t) - L \frac{di(t)}{dt} - \frac{1}{C} \int i(t) dt = 0 \quad \dots (1)$$

$$\text{Diff. (1), } +7i'(t) + i''(t) + 10i(t) = 0 \quad \dots (2)$$

Let $i''(t) = p^2$ and $i(t) = p$,

$$p^2 + 7p + 10 = 0$$

$$\therefore p = -5, -2$$

$\therefore$ The general solⁿ of above differential eqⁿ is

$$i(t) = k_1 e^{-2t} + k_2 e^{-5t} \quad \dots (3)$$

At $t=0^+$, $i(0^+) = k_1 + k_2$

$\Rightarrow k_1 + k_2 = 0 \quad \dots (4)$

And, differentiating (3), we get,

$i'(t) = -2k_1 e^{-2t} - 5k_2 e^{-5t} \quad \dots (5)$

At $t=0^+$, from (1), $20 - 7i(0^+) - i'(0^+) - 10 \int_0^0 i(t) dt = 0$
 $\Rightarrow 20 - 0 - i'(0^+) = 0$

$\therefore i'(0^+) = -20$

From (5), $2k_1 + 5k_2 = -20 \quad \dots (6)$

Solving (4) and (6), we get,

$k_1 + k_2 = 0 \quad \times 2$

$-2k_1 + 5k_2 = -20$

$-3k_2 = 20$

$\therefore k_2 = -6.67$ and $k_1 = 6.67$

$\therefore$ From (3), $i(t) = 6.67 e^{-2t} - 6.67 e^{-5t}$

$\Rightarrow$ Voltage across capacitor $(V_c(t)) = \frac{1}{C} \int i(t) dt$

$= \frac{1}{0.1} \left[\frac{6.67 e^{-2t}}{-2} - \frac{6.67 e^{-5t}}{-5} \right] + C$

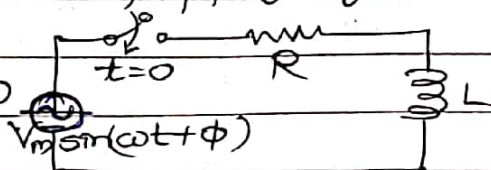
$= 10 \left[-3.335 e^{-2t} + 1.334 e^{-5t} \right] + C$

$\therefore V_c(t) = 13.34 e^{-5t} - 33.35 e^{-2t} + C$

Transient response of RL series ckt having sinusoidal excitation -

Let us consider a series RL-ckt with an alternating supplied voltage of $V_m \sin(\omega t + \phi)$. When the switch is closed at time $t=0$, applying KVL in the ckt,

$V_m \sin(\omega t + \phi) - i(t)R - L \frac{di(t)}{dt} = 0$



$$i_p(t) = e^{-\int R/L dt} \int \mathcal{Q} e^{\int R/L dt} dt$$

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$$\Rightarrow L i'(t) R + i(t) R = V_m \sin(\omega t + \phi)$$

$$\Rightarrow i'(t) + \frac{R}{L} i(t) = \frac{V_m}{L} \sin(\omega t + \phi) \quad \dots (1)$$

Comparing with standard form $\frac{dy}{dt} + Py = Q$,
 $P = \frac{R}{L}$ and $Q = \frac{V_m}{L} \sin(\omega t + \phi)$

$\therefore$ The solution of above differential eqⁿ is given by,
 $i(t) = e^{-\frac{R}{L}t} k + e^{-\frac{R}{L}t} \int \frac{V_m}{L} \sin(\omega t + \phi) e^{\frac{R}{L}t} dt$

where, $i_c(t) = k e^{-\frac{R}{L}t} \rightarrow$ complementary solⁿ

$i_p(t) = e^{-\frac{R}{L}t} \int \frac{V_m}{L} \sin(\omega t + \phi) e^{\frac{R}{L}t} dt \rightarrow$ particular solⁿ

$$\text{Now, } i_p(t) = \frac{V_m}{L} e^{-\frac{R}{L}t} \int \left(\frac{e^{j(\omega t + \phi)} - e^{-j(\omega t + \phi)}}{2j} \right) e^{\frac{R}{L}t} dt$$

$$= \frac{V_m e^{-\frac{R}{L}t}}{2Lj} \int e^{j(\omega t + \phi) + \frac{R}{L}t} - e^{-j(\omega t + \phi) + \frac{R}{L}t} dt$$

$$= \frac{V_m e^{-\frac{R}{L}t}}{2Lj} \left[\frac{e^{j(\omega t + \phi) + \frac{R}{L}t}}{(j\omega + \frac{R}{L})} - \frac{e^{-j(\omega t + \phi) + \frac{R}{L}t}}{(-j\omega + \frac{R}{L})} \right]$$

$$= \frac{V_m e^{-\frac{R}{L}t}}{2Lj \left(\frac{R^2}{L^2} + \omega^2 \right)} \left[\frac{e^{j(\omega t + \phi) + \frac{R}{L}t}}{(j\omega + \frac{R}{L})} - \frac{e^{-j(\omega t + \phi) + \frac{R}{L}t}}{(-j\omega + \frac{R}{L})} \right]$$

$$= \frac{V_m e^{-\frac{R}{L}t}}{2Lj} \left[\frac{2j}{\left(\frac{R^2}{L^2} + \omega^2 \right)} \left(\frac{e^{j(\omega t + \phi) + \frac{R}{L}t}}{(j\omega + \frac{R}{L})} - \frac{e^{-j(\omega t + \phi) + \frac{R}{L}t}}{(-j\omega + \frac{R}{L})} \right) \right]$$

$$= \frac{V_m e^{-\frac{R}{L}t}}{2L \left(\frac{R^2}{L^2} + \omega^2 \right)} \left[\frac{R \sin(\omega t + \phi) + j\omega \cos(\omega t + \phi)}{L} \right]$$

$$= \frac{V_m}{(R^2 + \omega^2 L^2)} [R \sin(\omega t + \phi) - \omega L \cos(\omega t + \phi)]$$

This can be reduced to sinusoidal form putting,

$$R = C \cos \theta \text{ and } \omega L = C \sin \theta.$$

$$\therefore C = \sqrt{R^2 + \omega^2 L^2} \text{ and } \theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Substituting this value,

$$i(t) = \frac{V_m}{R^2 + \omega^2 L^2} [C \sin(\omega t + \phi + \theta)]$$

$$= \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left\{ \sin\left(\omega t + \phi + \tan^{-1}\left(\frac{\omega L}{R}\right)\right) \right\}$$

$$\therefore i(t) = k e^{-R/L t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin\left(\omega t + \phi + \tan^{-1}\left(\frac{\omega L}{R}\right)\right)$$

Since inductor acts as open circuit at the time of switching, $i(0^+) = 0$.

$$\Rightarrow 0 = k + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin\left[\phi + \tan^{-1}\left(\frac{\omega L}{R}\right)\right]$$

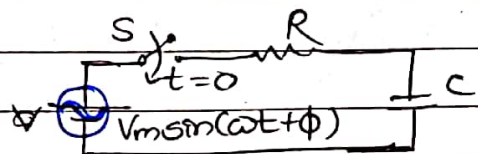
$$\therefore k = -\frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin\left(\phi + \tan^{-1}\left(\frac{\omega L}{R}\right)\right)$$

$$\therefore i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin(\omega t + \phi + \theta) - \sin(\phi + \theta) e^{-Rt/L} \right]$$

Transient response of series RC-circuit having sinusoidal excitation -

Consider a series RC ckt with alternating supplied

voltage $V_m \sin(\omega t + \phi)$. When the switch is closed at time $t=0$, applying KVL in the ckt,



$$V_m \sin(\omega t + \phi) - i(t)R - \frac{1}{C} \int i(t) dt = 0$$

$$\Rightarrow i(t)R + \frac{1}{C} \int i(t) dt = V_m \sin(\omega t + \phi)$$

Differentiating w.r.t t , we get,

$$i'(t)R + i(t) \frac{1}{C} = \omega V_m \cos(\omega t + \phi)$$

$$\Rightarrow i'(t) + i(t) \frac{1}{RC} = \frac{V_m \omega \cos(\omega t + \phi)}{R} \quad \dots (1)$$

This is non-homogeneous eqⁿ. Therefore, the current consists of sum of complementary function $i_c(t)$ and particular integral $i_p(t)$.

$$\text{i.e. } i(t) = i_c(t) + i_p(t)$$

$$\text{Here, } i_c(t) = e^{-\frac{t}{RC}}$$

$$\text{And, } i_p(t) = e^{-\frac{t}{RC}} \int \frac{V_m \omega \cos(\omega t + \phi)}{R} e^{\frac{t}{RC}} dt$$

$$= \frac{V_m \omega}{R} e^{-\frac{t}{RC}} \int \frac{e^{j(\omega t + \phi)} + e^{-j(\omega t + \phi)}}{2} \cdot e^{\frac{t}{RC}} dt$$

$$= \frac{V_m \omega}{2R} e^{-\frac{t}{RC}} \int e^{j(\omega t + \phi) + \frac{t}{RC}} + e^{-j(\omega t + \phi) + \frac{t}{RC}} dt$$

$$= \frac{V_m \omega}{2R} e^{-\frac{t}{RC}} \left[\frac{e^{j(\omega t + \phi) + \frac{t}{RC}}}{(j\omega + \frac{1}{RC})} + \frac{e^{-j(\omega t + \phi) + \frac{t}{RC}}}{(-j\omega + \frac{1}{RC})} \right]$$

$$= \frac{V_m \omega}{2R} \left\{ \frac{e^{j(\omega t + \phi)} (-j\omega + \frac{1}{RC}) - e^{-j(\omega t + \phi)} (j\omega + \frac{1}{RC})}{j\omega (\frac{1}{RC^2} + \omega^2)} \right\}$$

$$= \frac{V_m \omega}{2R} \left[-j\omega e^{j(\omega t + \phi)} + \frac{1}{RC} e^{j(\omega t + \phi)} - j\omega e^{-j(\omega t + \phi)} - \frac{1}{RC} e^{-j(\omega t + \phi)} \right]$$

$$\left(\omega^2 + \frac{1}{RC^2} \right)$$

$$= \frac{V_m \omega}{2R(\omega^2 + \frac{1}{RC})} \cdot \left[\frac{1}{RC} \{ e^{j(\omega t + \phi)} - e^{-j(\omega t + \phi)} \} - j\omega \{ \frac{e^{j(\omega t + \phi)}}{+e^{-j(\omega t + \phi)}} \} \right]$$

$$= \frac{V_m \omega}{2R(\omega^2 + \frac{1}{RC})} \cdot \left[\frac{1}{RC} \{ e^{j(\omega t + \phi)} - e^{-j(\omega t + \phi)} \} - j\omega \{ \frac{e^{j(\omega t + \phi)}}{+e^{-j(\omega t + \phi)}} \} \right]$$

$$= \frac{V_m \omega}{2R(\omega^2 + \frac{1}{RC})} \cdot \left[\frac{2j}{RC} \sin(\omega t + \phi) - 2j\omega \cos(\omega t + \phi) \right]$$

$$= \frac{V_m \omega j}{R\omega^2 RC + R} \left[\sin(\omega t + \phi) - \omega RC \cos(\omega t + \phi) \right]$$

The Laplace transform is one of the mathematical tool used for the solution of linear ordinary integro-differential equation. In comparison with the classical method, Laplace has following features -

1. The homogeneous equation and particular integral of solution are obtained in one operation.
2. The Laplace transform converts integro-differential equation into an algebraic form in Laplace operator (s). It is then possible to manipulate algebraic equation by simple algebraic rules and the final solution can be obtained by taking inverse transform. The Laplace transform method is a powerful technique or tool for solving circuits containing R, L or C.

For the time function $f(t)$, the Laplace transform is given by

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

where

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1. $1, k$	$\frac{1}{s}, \frac{k}{s}$
2. t, t^n	$\frac{1}{s^2}, \frac{n!}{s^{n+1}}$
3. e^{at}	$\frac{1}{s-a}$
4. e^{-at}	$\frac{1}{s+a}$
5. te^{iat}	$\frac{1}{(s+a)^2}$

6. $\sin \omega t$

$$\frac{\omega}{s^2 + \omega^2}$$

7. $\cos \omega t$

$$\frac{s}{s^2 + \omega^2}$$

$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$

$\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0)$

$\int \mathcal{L}\{f(t)\} dt = \frac{F(s)}{s}$

Q.1. For the given circuit, find the expression for the current $i(t)$ when switch is closed at $t=0$ assuming the initial current through inductor is 3A.

sol) Applying KVL when switch is closed at $t=0$,

$$40 - 4i(t) - \frac{d i(t)}{dt} = 0$$

$$\Rightarrow i(t) + 4i(t) - 40 = 0$$

Taking Laplace transform on both sides,

$$sI(s) - \overset{3}{I(0)} + 4I(s) - \frac{40}{s} = 0$$

$$\Rightarrow I(s) [s+4] = \frac{40}{s} + 3$$

$$\therefore I(s) = \frac{40+3s}{s(s+4)}$$

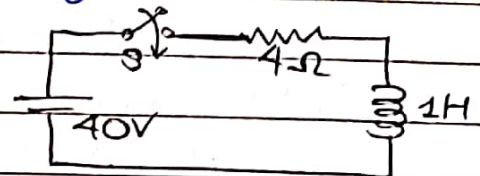
$$\text{Let } \frac{40+3s}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

$$\Rightarrow A = \frac{40+3s}{s+4} \Big|_{s=0} \therefore A = 10$$

$$\text{And, } \frac{B}{s} = \frac{40+3s}{s} \Big|_{s=-4} = -7$$

$$\therefore I(s) = \frac{10}{s} - \frac{7}{s+4}$$

Taking inverse Laplace transform on both sides,

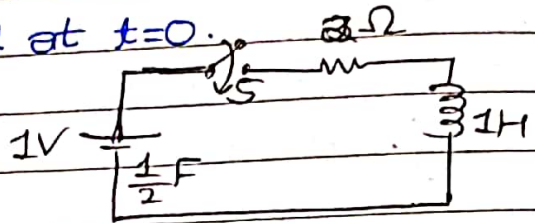


$$\Rightarrow i(s) = 10 u(t) - 7 e^{-4t}$$

which is the required solution of current.

2. Find the expression for current across inductor when switch is closed at $t=0$.

sol) Applying KVL in the ckt when switch is closed at $t=0$,



$$\frac{1}{\left(\frac{1}{2}\right)} \int i(t) dt - 2 i(t) - \frac{d i(t)}{dt} = 0 \quad \dots (1)$$

*) Differentiating w.r.t t on both sides,

$$2 i(t) + 2 i'(t) + i''(t) = 0$$

Taking Laplace transform on both sides, we get,

$$\frac{1}{s} - \frac{2 I(s)}{s} - 2 I(s) - \{s I(s) - I(0)\} = 0$$

$$\Rightarrow \frac{1}{s} - \frac{2 I(s)}{s} - 2 I(s) - s I(s) + I(0) = 0$$

$$\Rightarrow \frac{1}{s} - I(s) \left(\frac{2}{s} + 2 + s \right) = 0$$

$$\Rightarrow \frac{1}{s} - I(s) \left(\frac{s^2 + 2s + 2}{s} \right) = 0$$

$$\Rightarrow I(s) = \frac{1}{(s^2 + 2s + 1) + 1} = \frac{1}{(s+1)^2 + 1^2}$$

Taking Laplace inverse on both sides,

$$i(t) = e^{-t} \cdot \sin t$$

which is the required solution.

Q1. Solve the following differential eq: $i'' + 3i' + 2i = 0$ given that $i(0) = 2$, $i'(0) = -3$ using Laplace transform

sol The given differential equation is $i'' + 3i' + 2i = 0$

Taking Laplace transform on both sides,

$$s^2 I(s) - sI(0) - I'(0) + 3[sI(s) - I'(0)] + 2I(s) = 0$$

$$\Rightarrow s^2 I(s) - 2s + 3 + 3sI(s) - 3 + 2I(s) = 0$$

$$\Rightarrow (s^2 + 3s + 2) I(s) - 2s = 0$$

$$\Rightarrow I(s) = \frac{2s+3}{s^2+3s+2} = \frac{2s+3}{(s+2)(s+1)}$$

$$\text{Let } \frac{2s+3}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$\text{Here, } \frac{A}{(s+2)} = \frac{2s+3}{s+2} \Big|_{s=-1} = 1$$

$$B = \frac{2s+3}{s+1} \Big|_{s=-2} = 1$$

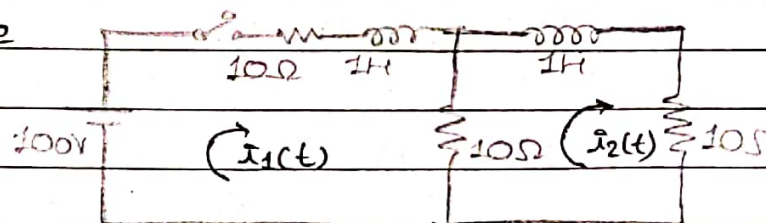
$$\therefore I(s) = \frac{1}{(s+1)} + \frac{1}{(s+2)}$$

Taking Laplace inverse on both sides,

$$i(t) = e^{-t} + e^{-2t}$$

2. In the given circuit, switch is closed at $t=0$, find the expression of current $i_1(t)$ and $i_2(t)$ if the circuit is unenergized before switch is closed.

sol Applying KVL in the loop (1),



$$100 - 10i_1(t) - i_1'(t) - 10[i_1(t) - i_2(t)] = 0$$

$$\Rightarrow 100 - 10i_1(t) - i_1'(t) - 10i_1(t) + 10i_2(t) = 0$$

$$\Rightarrow 20i_1(t) + i_1'(t) - 10i_2(t) = 100 = 0 \quad \text{--- (1)}$$

Taking Laplace transform on both sides,

$$20I_1(s) + sI_1(s) - I_1(0) - 10I_2(s) - 100 = 0$$

$$\Rightarrow 20I_1(s) + sI_1(s) - I_1(0) - 10I_2(s) = \frac{100}{s} = 0$$

$$\Rightarrow (s^2 + 20s)I_1(s) - 10I_2(s) = \frac{100}{s} \quad \dots (2)$$

Applying KVL in loop (2), we get,

$$-10[i_2(t) - i_1(t)] - i_2'(t) - 10i_2(t) = 0$$

$$\Rightarrow -10i_2(t) + 10i_1(t) - i_2'(t) - 10i_2(t) = 0$$

$$\Rightarrow -20i_2(t) + 10i_1(t) - i_2'(t) = 0$$

Taking Laplace inverse on both sides,

$$-20I_2(s) + 10I_1(s) - sI_2(s) - I_2(0) = 0$$

$$\Rightarrow 10I_1(s) - (20+s)I_2(s) = 0$$

$$\Rightarrow I_2(s) = \frac{10I_1(s)}{20+s} \quad \dots (3)$$

Substituting this value in (2),

$$(s^2 + 20s)I_1(s) - 10 \cdot \frac{10I_1(s)}{20+s} = \frac{100}{s}$$

$$\Rightarrow (s^2 + 400s + 400 - 100)I_1(s) = \frac{100(20+s)}{s}$$

$$\Rightarrow (s^2 + 400s + 300)I_1(s) = \frac{100(20+s)}{s}$$

$$\Rightarrow I_1(s) = \frac{100(20+s)}{s(s+30)(s+10)}$$

$$\text{Let } \frac{100(20+s)}{s(s+30)(s+10)} = \frac{A}{s} + \frac{B}{s+30} + \frac{C}{s+10}$$

$$\text{Here, } A = \frac{100(20+s)}{(s+30)(s+10)} \Big|_{s=0}$$

$$\therefore A = \frac{20}{3}$$

$$B = \frac{100(20+s)}{s(s+10)} \Big|_{s=-30} = -\frac{5}{3}$$

$$C = \frac{100(20+s)}{s(s+30)} \Big|_{s=-10} = -5$$

$$\therefore I_1(s) = \frac{100 \cdot 20}{3s} - \frac{5}{3(s+30)} - \frac{5}{(s+10)}$$

$$\therefore i_1(t) = \frac{20}{3} u(t) - \frac{5}{3} e^{-30t} - 5 e^{-10t}$$

$$\text{From (1), } (s+20) I_1(s) - 10 I_2(s) = \frac{100}{s}$$

$$\Rightarrow \frac{(s+20) \cdot (s+20)}{10} I_2(s) - 10 I_2(s) = \frac{100}{s}$$

$$\Rightarrow \frac{(s^2 + 400s + 400 - 100)}{10} I_2(s) = \frac{100}{s}$$

$$\Rightarrow (s^2 + 400s + 300) I_2(s) = \frac{1000}{s}$$

$$\therefore I_2(s) = \frac{1000}{s(s+30)(s+10)}$$

$$\text{Let } \frac{1000}{s(s+30)(s+10)} = \frac{A}{s} + \frac{B}{s+30} + \frac{C}{s+10}$$

$$\text{Here, } A = \frac{1000}{(s+30)(s+10)} \Big|_{s=0} = \frac{10}{3}$$

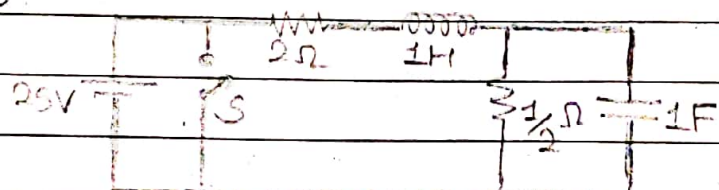
$$B = \frac{1000}{s(s+10)} \Big|_{s=-30} = \frac{5}{3}$$

$$C = \frac{1000}{s(s+30)} \Big|_{s=-10} = -5$$

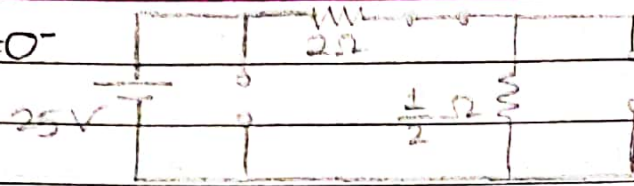
$$\therefore I_2(s) = \frac{10}{3s} + \frac{5}{3(s+30)} - \frac{5}{(s+10)}$$

$$\therefore i_2(t) = \frac{10}{3} u(t) + \frac{5}{3} e^{-30t} - 5 e^{-10t}$$

3. In the given ckt, steady state condⁿ is raised when switch is opened, determine the current $i(t)$ and voltage across capacitor at $t > 0$ when switch is closed.

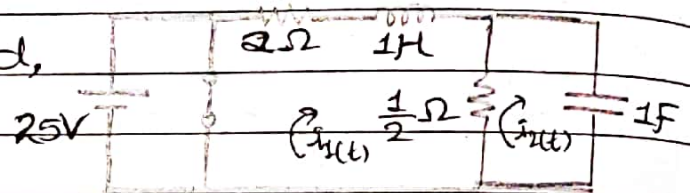


sol

At $t=0^-$ 

Here, $i(0^-) = \frac{25}{\left(\frac{5}{2}\right)} = 10 \text{ A}.$

When switch is closed,

Applying KVL in
loop (1),

$$-2i_1(t) - \dot{i}_1(t) - \frac{1}{2}i_1(t) + \frac{1}{2}i_2(t) = 0$$

$$\Rightarrow \frac{1}{2}i_2(t) - \frac{5}{2}i_1(t) - \dot{i}_1(t) = 0 \quad \dots (1)$$

Taking Laplace transform on both sides,

Response of RL series circuit using Laplace Transform.

Applying KVL in the ckt,

$$V - V_R - V_L = 0$$

$$\Rightarrow V - i(t)R - L \frac{di(t)}{dt} = 0 \quad \dots (1)$$

Taking Laplace transform,

$$\frac{V}{s} - RI(s) - L\{sI(s) - I(0)\} = 0$$

$$\Rightarrow \frac{V}{s} - RI(s) - LsI(s) = 0$$

$$\Rightarrow sI(s) (R + L) = \frac{V}{s}$$

$$\therefore I(s) = \frac{V}{s(R + L)} = \frac{V}{sL(R/L + 1)} = \frac{A}{s} + \frac{B}{(R/L + s)}$$

$$\text{Here, } \frac{VA}{sL(R/L + 1)(R/L + s)} \Big|_{s=0} = \frac{V}{R}$$

$$B = \frac{V}{sL} \Big|_{s=-\frac{R}{L}} = -\frac{V}{R}$$

$$\therefore I(s) = \frac{V}{Rs} - \frac{V}{R(R/L + s)}$$

Taking Laplace inverse on both sides,

$$i(t) = \frac{V}{R} [1 - e^{-R/L t}]$$

Response of RC

Applying KVL in the ckt,

$$V - V_R - V_C = 0$$

$$\Rightarrow V - i(t)R - \frac{1}{C} \int i(t) dt = 0$$

Taking Laplace transform on both sides,

$$\frac{V}{s} - RI(s) - \frac{1}{C} \cdot \frac{I(s)}{s} = 0$$

$$\Rightarrow \frac{V}{s} I(s) \left(R + \frac{1}{Cs} \right) = \frac{V}{s}$$

$$\Rightarrow I(s) = \frac{V}{s \left(\frac{RCs + 1}{Cs} \right)} = \frac{VC}{(RCs + 1)} = \frac{VC}{\left(s + \frac{1}{RC} \right) RC}$$

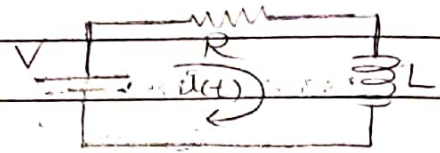
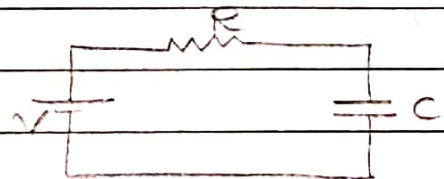


fig - RL series ckt



$$\Rightarrow I(s) = \frac{V}{R(s + 1/RC)}$$

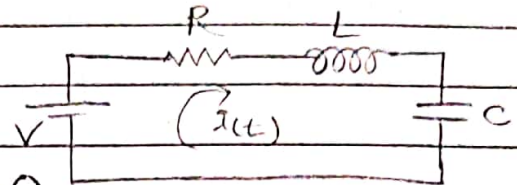
Taking Laplace inverse on both sides,

$$i(t) = \frac{V}{R} e^{-R/C t}$$

Response of RLC-

Applying KVL in the ckt,

$$V - V_R - V_C = V_L = 0$$



$$\Rightarrow V - i(t)R - L \frac{di(t)}{dt} - \frac{1}{C} \int i(t) dt = 0$$

Taking Laplace transform on both sides,

$$\frac{V}{s} - R I(s) - L \{s I(s) - I(0)\} - \frac{1}{C} \cdot \frac{I(s)}{s} = 0$$

$$\Rightarrow \frac{V}{s} - I(s) \left(R + Ls + \frac{1}{Cs} \right) = 0$$

$$\Rightarrow I(s) = \frac{V}{s(RCs + Ls^2 + 1)} = \frac{VC}{(RCs + Ls^2 + 1)}$$

$$\text{Then, } s^2 LC + RCs + 1 = 0$$

$$\therefore s = \frac{-RC \pm \sqrt{R^2 C^2 - 4LC}}{2LC} = \frac{-R}{2L} \pm \sqrt{\frac{R^2 C^2 - 4LC}{2LC}}$$

Let k_1 and k_2 be the two roots of s .

$$\text{Then, } I(s) = \frac{VC}{(s - k_1)(s - k_2)} = \frac{A}{(s - k_1)} + \frac{B}{(s - k_2)}$$

$$\Rightarrow I(s) \approx \text{Here, } A = \frac{VC}{s - k_2} \Big|_{s=k_1} = \frac{VC}{k_1 - k_2}$$

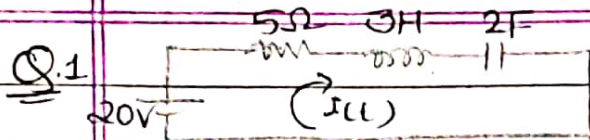
$$B = \frac{VC}{s - k_1} \Big|_{s=k_2} = \frac{VC}{k_2 - k_1}$$

$$\therefore I(s) = \frac{VC}{(s - k_1)(k_1 - k_2)} + \frac{VC}{(s - k_2)(k_2 - k_1)}$$

$$\Rightarrow I(s) = VC \left[\frac{1}{(s - k_1)(k_1 - k_2)} + \frac{1}{(s - k_2)(k_2 - k_1)} \right]$$

Taking Laplace inverse on both sides,

$$i(t) = VC \left\{ \frac{e^{k_1 t}}{k_1 - k_2} + \frac{e^{k_2 t}}{k_2 - k_1} \right\}$$



for Applying KVL in the ckt, $20 - 5i(t) - 3\frac{di(t)}{dt} - \frac{1}{2}\int i(t)dt = 0$

Taking Laplace transform on both sides,

$$\frac{20}{s} - 5I(s) - 3\{sI(s) - i(0)\} - \frac{1}{2} \cdot \frac{I(s)}{s} = 0$$

$$\Rightarrow \frac{20}{s} - 5I(s) - 3sI(s) - \frac{I(s)}{2s} = 0$$

$$\Rightarrow I(s) \left(5 + 3s + \frac{1}{2s} \right) = \frac{20}{s}$$

$$\Rightarrow I(s) \left(\frac{6s^2 + 10s + 1}{2s} \right) = \frac{20}{s}$$

$$\Rightarrow I(s) = \frac{40}{6s^2 + 10s + 1}$$

Now, for roots: $6s^2 + 10s + 1 = 0 \therefore k_1 = -0.106$

$$\therefore I(s) = \frac{40}{(s+0.106)(s+1.56)} = \frac{A}{(s+0.106)} + \frac{B}{(s+1.56)}$$

$$\text{Here, } A = \frac{40}{s+1.56} \Big|_{s=-0.106} = 27.5$$

$$B = \frac{40}{s+0.106} \Big|_{s=-1.56} = -27.5$$

$$\therefore I(s) = \frac{27.5}{(s+0.106)} - \frac{27.5}{(s+1.56)}$$

Taking Laplace inverse,

$$i(t) = 27.5 \left[e^{-0.106t} - e^{-1.56t} \right]$$

Initial value theorem - If $f(t)$ and $f'(t)$ are Laplace transformable, then the initial value $f(0)$ is given by $f(0) = \lim_{s \rightarrow \infty} sF(s)$.

Final value theorem - If $f(t)$ and $f'(t)$ are Laplace transformable, then the final value $f(\infty)$ is given by $f(\infty) = \lim_{s \rightarrow 0} sF(s)$

Transfer functions -

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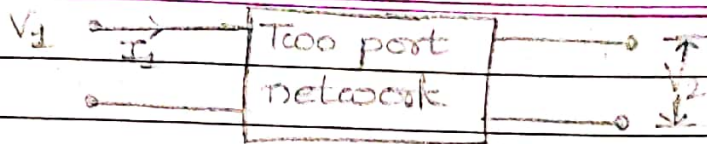


fig - Two port network

Transfer function of a system is defined as the ratio of Laplace transform of o/p to Laplace transform of i/p. The transfer function is used to describe the stability of network of circuit which have atleast two ports - i/p & o/p port.

The transfer function relates the transform of quantity from one port to another port. In electrical ckt, there are two major quantities i.e voltage and current. The ratio of o/p voltage to i/p voltage is known as voltage transfer function/ratio. i.e $V_{TF} = V_{TR} = \frac{V_{out}}{V_{in}} = \frac{V_2}{V_1}$

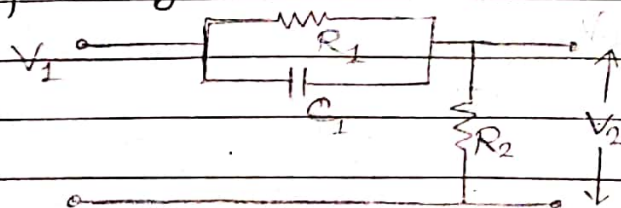
The ratio of o/p current to i/p current is known as current transfer function i.e $I_{TF} = \frac{I_{out}}{I_{in}} = \frac{I_2}{I_1}$

The ratio of o/p voltage to i/p current is known as impedance transfer function i.e $Z_{TF} = \frac{V_o}{I_i} = \frac{V_2}{I_1}$

The ratio of o/p current to i/p current is known as admittance transfer function. i.e $Y_{TF} = \frac{I_2}{V_1}$

Q.1. Determine the transfer function $\frac{V_2}{V_1}$ for given ckt.

solⁿ Replacing the ckt

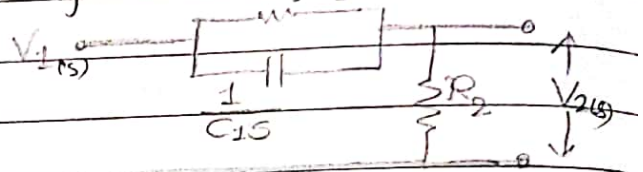


Ex/ Replacing the ckt using s operator, R_1

$$Z_1 = R_1 \parallel \frac{1}{C_1 s}$$

$$= \frac{R_1 / C_1 s}{(R_1 C_1 s + 1) / C_1 s}$$

$$\therefore Z_1 = \frac{R_1}{R_1 C_1 s + 1}$$

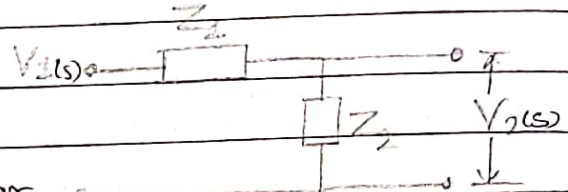


$$Z_2 = R_2$$

(Applying voltage divider formula,

$$V_2(s) = V_1(s) \times \frac{Z_2}{Z_1 + Z_2}$$

$$\Rightarrow \frac{V_2(s)}{V_1(s)} = \frac{R_2}{R_2 + \left(\frac{R_1}{R_1 C_1 s + 1} \right)}$$

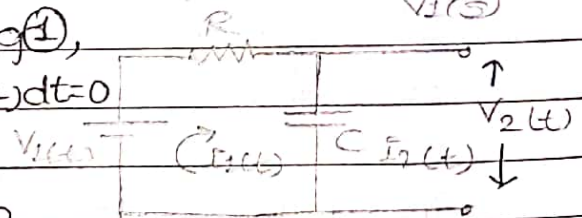


$$\Rightarrow \frac{V_2(s)}{V_1(s)} = \frac{R_2(R_1 C_1 s + 1)}{R_1 + R_2 + R_1 R_2 C_1 s}$$

2. Determine the voltage transfer function $V_2(s)$.

sol/ Applying KVL in the loop (1),

$$V_1(t) - R i_1(t) - \frac{1}{C} \int i_1(t) - i_2(t) dt = 0$$



Taking Laplace transform on both sides,

$$V_1(s) - R I_1(s) - \frac{1}{C} [I_1(s) - I_2(s)] = 0$$

$$\Rightarrow V_1(s) - I_1(s) \left(R + \frac{1}{C s} \right) = 0$$

$$\Rightarrow I_1(s) = I_2(s) \left(\frac{R C s + 1}{C s} \right) \quad \dots (1)$$

Applying KVL in loop (2), we get,

$$V_2(t) = \frac{1}{C} \int i_2(t) - i_1(t) dt$$

Taking Laplace transform on both sides,

$$V_2(s) = -\frac{I_1(s)}{C s} \quad \dots (2)$$

Dividing eqn (2) by (1), we get, $\frac{V_2(s)}{V_1(s)} = \frac{-I_1(s) \cdot Cs}{Cs \cdot I_1(s) \downarrow (R_1 + 1)}$
 $\therefore \frac{V_2(s)}{V_1(s)} = \frac{-1}{R_1 + 1}$

3. Find $\frac{V_2(s)}{V_1(s)}$ for the given ckt. $s/s+2$ $s/s+2$

Applying KVL in loop (1), $V_1(s) - \frac{s}{s+2} I_1(s) - \frac{1}{s+1} \{I_1(s) - I_2(s)\} = 0$

$$\Rightarrow V_1(s) - \frac{s}{s+2} I_1(s) - \frac{1}{s+1} I_1(s) + \frac{I_2(s)}{s+1} = 0$$

$$\Rightarrow V_1(s) = I_1(s) \left\{ \frac{s}{s+2} + \frac{1}{s+1} \right\} - \frac{I_2(s)}{s+1}$$

$$= I_1(s) \left\{ \frac{s^2 + s + s + 2}{(s+1)(s+2)} \right\} - \frac{I_2(s)}{s+1}$$

$$\therefore V_1(s) = \frac{(s^2 + 2s + 2)}{(s+1)(s+2)} I_1(s) - \frac{I_2(s)}{s+1} \quad \dots (1)$$

Applying KVL in loop (2), we get,

$$-\frac{1}{s+3} I_2(s) + \frac{1}{s+3} I_1(s) - \frac{s}{s+2} I_2(s) - \frac{I_2(s)}{s+3} = 0$$

$$\Rightarrow \frac{I_1(s)}{s+3} - I_2(s) \left\{ \frac{1}{s+3} + \frac{s}{s+2} + \frac{1}{s+3} \right\} = 0$$

$$\Rightarrow \frac{I_1(s)}{s+3} = I_2(s)$$

Calculation of network function for ladder network

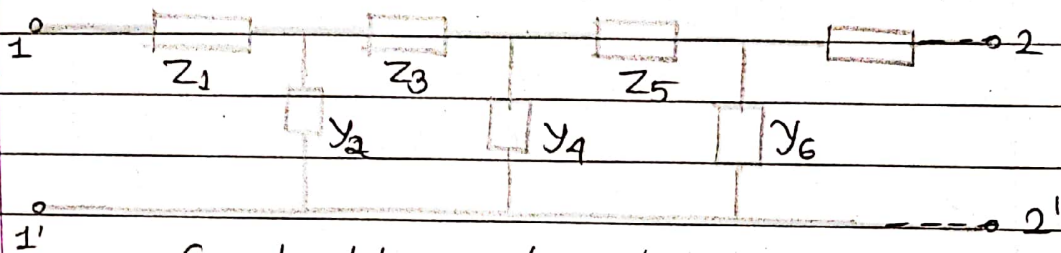


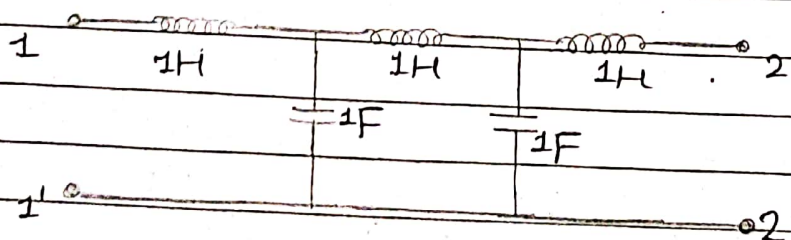
fig - Ladder network

The driving point impedance of two port network at i/p terminal is given by,

$$Z_{11'} = \frac{1}{Z_1 + \frac{1}{Y_2 + \frac{1}{Z_3 + \frac{1}{Y_4 + \dots}}}}$$

It is also known as continued fraction. It may be simplified to determine the driving point impedance for the given network.

Q. Determine the driving point impedance for the given network:-



Sol

Replacing the circuit using s-operator,

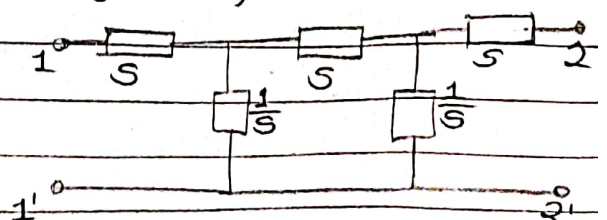
Here, $Z_1 = s$

$Y_2 = s$

$Z_3 = s$

$Y_4 = s$

$Z_5 = s$

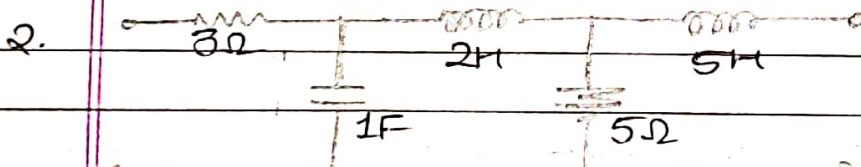


$$Z_{11}' = \frac{1}{s + \frac{1}{s + \frac{1}{s + \frac{1}{s}}}}$$

$$= \frac{1}{111} = \frac{5 + 1}{5 + 1} = \frac{5 + 1}{5 + 15} = \frac{5 + 1}{5^2 + 1}$$

$$= \frac{\cancel{1}}{5 + \frac{1}{5 + \frac{(1s^2+1)}{s^3+2s}}}} = \frac{\cancel{1}}{5 + \frac{-s^3+2s}{s^4+3s^2+1}}}$$

$$\therefore \overline{Z_{q1}}' = \frac{1}{s^5 + 4s^3 + 3s} \quad \therefore Z_{11}' = \frac{s^4 + 3s^2 + 1}{s^5 + 4s^3 + 3s}$$



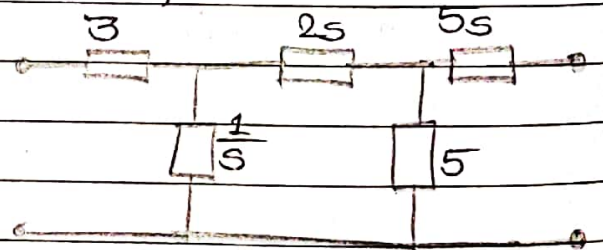
Here, $R_1 = 3$

$$\underline{Y_2 = 5}$$

$$Z_3 = 25$$

$$Z_1 = \frac{1}{5}$$

$$Z_5 = 55$$


$$Z_{eq} = Z_1 + \frac{1}{\frac{1}{Y_2 + \frac{1}{Z_3 + \frac{1}{Y_4 + \frac{1}{Z_5}}}}}$$

$$= 3 + 1$$

$$6 + 1$$

$$2s + 1$$

$$\frac{5}{5} + \frac{1}{5s}$$

$$= 3 + 1$$

$$s + 1$$

$$2s + 5s$$

$$2(s+1)$$

$$= 3 + 1$$

$$s + 5s(s+1)$$

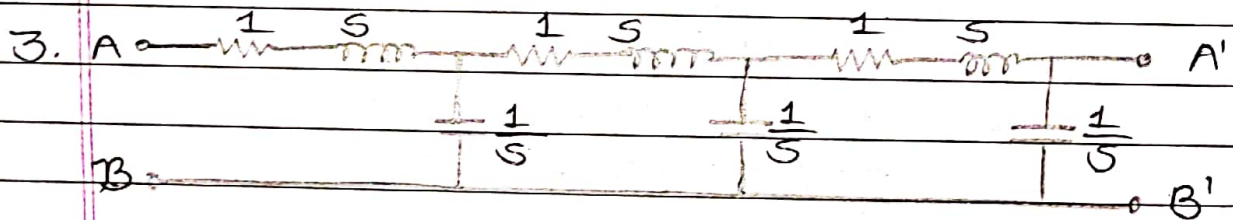
$$10s^2 + 5s + 1$$

$$= 3 + \frac{2s^2 + 7s}{2s^3 + 7s^2 + s + 1}$$

$$2s^3 + 7s^2 + s + 1$$

$$\therefore Z_{eq} = \frac{6s^3 + 23s^2 + 10s + 3}{2s^3 + 7s^2 + s + 1}$$

$$2s^3 + 7s^2 + s + 1$$



Sol: Replacing the elt using s-operator

Here, $Z_1 = s+1$

$$Y_2 = \frac{1}{s}$$

$$Z_3 = s+1$$

$$Y_4 = \frac{1}{s}$$

$$Z_5 = s+1, Y_6 = \frac{1}{s}$$

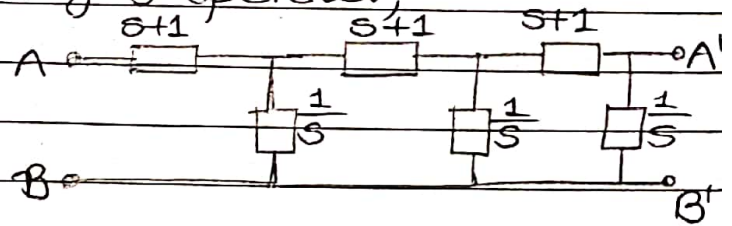
$$\therefore Z_{eq} = Z_1 + \frac{1}{Y_2 + \frac{1}{Z_3 + \frac{1}{Y_4 + \frac{1}{Z_5 + \frac{1}{Y_6}}}}}$$

$$Y_2 + \frac{1}{Z_3 + \frac{1}{Y_4 + \frac{1}{Z_5 + \frac{1}{Y_6}}}}$$

$$Z_3 + \frac{1}{Y_4 + \frac{1}{Z_5 + \frac{1}{Y_6}}}$$

$$Y_4 + \frac{1}{Z_5 + \frac{1}{Y_6}}$$

$$Z_5 + \frac{1}{Y_6}$$



$$= s+1 + \frac{1}{s+1}$$

$$s+1$$

$$s+1 + \frac{1}{s+1}$$

$$s+1$$

$$(s+1) + \frac{1}{s}$$

$$= s+1 + \frac{1}{s+1}$$

$$s+1$$

$$s+1 + \frac{1}{s+1}$$

$$s+1$$

$$s^2+s+1$$

$$= (s+1) + \frac{1}{s+1}$$

$$s+1$$

$$(s+1) + \left(\frac{s^2+s+1}{s^3+s^2+2s} \right)$$

$$= (s+1) + \frac{1}{s+1}$$

$$s+1$$

$$s^4+s^3+2s^2+s^3+2s^2+2s$$

$$= (s+1) + \frac{s^4+s^3+2s^2+s^3+2s^2+2s}{s^5+2s^4+4s^3+2s^2+s^3+s^2+2s}$$

$$s^5+2s^4+4s^3+2s^2+s^3+s^2+2s$$

$$= (s+1) + \frac{s^4+s^3+2s^2+2s}{s^5+5s^3+2s^4+3s^2+2s}$$

$$s^5+5s^3+2s^4+3s^2+2s$$

$$= \frac{(s^6+5s^4+2s^5+3s^3+2s^2+s^5+5s^3+2s^4+3s^2+2s+s^4+2s^3+4s^2+2s)}{s^5+5s^3+2s^4+3s^2+2s}$$

$$+4s^2+2s$$

$$= \frac{s^6+3s^5+7s^4+10s^3+9s^2+4s}{s^5+5s^3+2s^4+3s^2+2s}$$

$$\therefore Z_{eq} = \frac{s^6+3s^5+7s^4+10s^3+9s^2+4s}{s^5+5s^3+2s^4+3s^2+2s}$$

$$s^5+5s^3+2s^4+3s^2+2s$$

Poles and zeroes of network func-

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The transfer function $T(s)$ are the rational function of s -operator and may be expressed as the ratio of two polynomials, numerator $N(s)$ and denominator $D(s)$. The polynomial $N(s)=0$ has n no^s of roots and they are known as zeroes of network function and the denominator $D(s)=0$ has m no^s of roots and they are known as poles of network function

$$\therefore T(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_0}{b_m s^m + b_{m-1} s^{m-1} + b_{m-2} s^{m-2} + \dots + b_0}$$

$$\therefore T(s) = \frac{(s-z_1)(s-z_2)(s-z_3)\dots(s-z_m)}{(s-p_1)(s-p_2)(s-p_3)\dots(s-p_m)}$$

where, $z_1, z_2, z_3 \dots z_n$ are zeroes of transfer func.
 $p_1, p_2, \dots, p_m$ are poles of transfer func.

These are the required poles and zeroes of transfer function and their location in s -plane completely specify the network function. In complex s -plane, zeroes are represented by small circle and poles are represented by small cross.

Q.1 Obtain the pole, zero location for transfer function $T(s) = \frac{(2s+4)(s+4)}{s(s+1)(s+3)}$

~~Ans~~ Also plot the poles and zeroes in the imaginary complex s -plane.

Q1: Here, $T(s) = \frac{(2s+4)(s+4)}{s(s+1)(s+3)} = \frac{N(s)}{D(s)}$

For zeroes: $(2s+4)(s+4) = 0$

$\therefore s = -2, -4.$

For poles: $s(s+1)(s+3) = 0$

$\therefore s = 0, -1, -3$

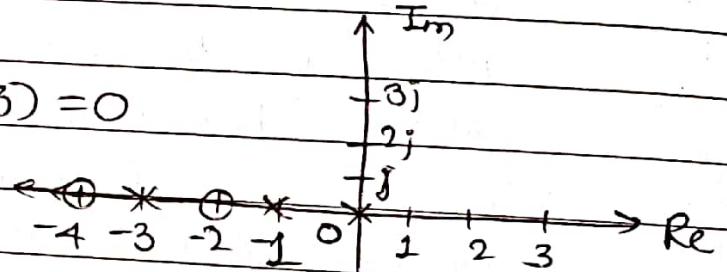


fig-zeroes and poles represent-
ation of transfer function

$H(s) = \frac{s(s+1)}{(s+4)(s^2+6s+18)}$

Q2: Here, $H(s) = \frac{s(s+1)}{(s+4)(s^2+6s+18)} = \frac{N(s)}{D(s)}$

For zeroes: $N(s) = 0$

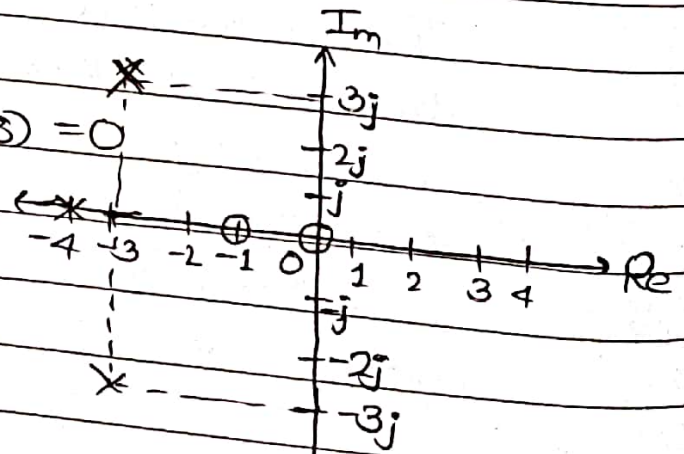
$\Rightarrow s(s+1) = 0$

$\therefore s = 0, -1.$

For poles: $D(s) = 0$

$\Rightarrow (s+4)(s^2+6s+18) = 0$

$\therefore s = -4, -3 \pm 3j$



Routh-Hurwitz criteria for the stability of T.F.:

The necessary conditions for the system to be stable is that the roots of the characteristic equation $D(s) = 0$ or poles of transfer function has negative real part.

Routh-Hurwitz stability criteria is the method for determining number of roots of the characteristic equation.

Consider characteristic eqn in the form,

$$D(s) = a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + a_{n-3} s^{n-3} + \dots + a_0$$

where, $a_n, a_{n-1}, \dots, a_0$ are the constants.

In order that, the above equation doesn't contain roots with positive real part, the following conditions are necessary -

1. All the coefficient of the equations should have same sign.
2. There should not be any term missing betⁿ the highest and lowest degree.

&

The necessary and sufficient condition that all the roots of above equation can be calculated from the Routh array.

If all the elements of first column of array are of same sign, then the system is stable otherwise it is unstable.

The no^o of changes of signs in the first column is equal to the no^o of roots with positive real parts.

Routh Tabulation/Array can be calculated as:

Let $D(s)$ be a polynomial of order '6' as,

$$D(s) = a_6 s^6 + a_5 s^5 + a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0$$

Routh array:-

s^6	a_6	a_4	a_2	a_0
s^5	a_5	a_3	a_1	0
s^4	$\frac{a_5 a_4 - a_6 a_3}{a_5} = A$	$\frac{a_5 a_2 - a_6 a_1}{a_5} = B$	$\frac{a_5 a_0 - a_6 \cdot 0}{a_5} = 0$	0
s^3	C	D	0	0
s^2	E	F	0	0
s^1	G	0	0	0
s^0	H	0	0	0

Here, $C = \frac{A a_3 - B a_5}{A}$, $D = \frac{A a_1 - a_5 a_0}{A}$

$E = \frac{B C - A D}{C}$, $F = \frac{C a_0 - a_5 a_0}{C}$

$G = \frac{E D - C F}{E}$, $H = \frac{G F}{G} = F = a_0$

Determine the stability of the system whose characteristic eqn is given by,

$$D(s) = s^4 + 8s^3 + 18s^2 + 16s + 5$$

✓ We have, $D(s) = s^4 + 8s^3 + 18s^2 + 16s + 5$

Routh Array can be calculated as,

s^4	1	18	5
s^3	8	16	0
s^2	16	5	0
s	13.5	0	0
s^0	5	0	0

All the terms in the first column of Routh array are positive. Hence the given system is stable.

Q. $D(s) = s^4 + 2s^3 + 3s^2 + 4s + 5$

sol. Routh Array can be calculated as,

s^4	1	3	5
s^3	2	4	0
s^2	1	5	0
s^1	-6	0	0
s^0	5	0	0

Since all the terms of first column of Routh array are not of same sign, the given system is not stable.

Q. $s^3 + 7s^2 + 10s + 10k = 0$

sol. Routh Array can be tabulated as,

s^3	1	10	0
s^2	7	$10k$	0
s^1	$\frac{70-10k}{7}$	0	0
s^0	$10k$	0	0

The given system is stable if $10k > 0$.

OR, $\frac{70-10k}{7} > 0$

$\Rightarrow 70 - 10k > 0$

$\therefore k < 7$

$\therefore$ Value of k is $0 < k < 7$

Here, we study the network function which describes the network in a sinusoidal steady state where $s = j\omega$.

Let $G(s)$ be any network function, then
 $G(s) =$

Cases of Hurwitz stability:

1. If the first column of Routh-Hurwitz array consists of 0 value, replace the zero by small positive quantity α and continue the process of Routh array.
2. When any one row of the Routh array is zero. In such case, the array can be completed by forming auxiliary equation from the previous row and determining the first derivative with s -operator, the required row can be determinant.

eg $P(s) = s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50$

⇒ The Routh array can be tabulated as:

s^5	1	24	-25
s^4	2	48	-50
s^3	0	0	0

Since the third row of Routh array consists of all zeroes, so the auxiliary equation of previous row is given by,

$$A(s) = 2s^4 + 48s^2 - 50$$

$$A'(s) = 8s^3 + 96s$$

Replacing third row with the coefficient of $A'(s)$, we get,

s^5	1	24	-25
s^4	2	48	-50
s^3	8	56	0
s^2	24	-50	0
s^1	112.67	0	0
s^0	-50	0	0

Since the first column consists of zero, the given system is unstable.

CHAPTER 7

Frequency Response of Network

Here, we study the network function which describes the network in a sinusoidal steady state where $s = j\omega$.

Let $G(s)$ be any network function, then

$G(s) = G(j\omega)$ is a complex function which can be written as $G(j\omega) = \text{Re}[G(j\omega)] + j\text{Im}[G(j\omega)]$

$$\Rightarrow |G(j\omega)| = \sqrt{(\text{Re}[G(j\omega)])^2 + (\text{Im}[G(j\omega)])^2}$$

$$\text{And, } \theta = \tan^{-1} \left(\frac{\text{Im}[G(j\omega)]}{\text{Re}[G(j\omega)]} \right)$$

Bode plot: It is a graphical representation of the transfer function for determining the stability of the system. It consists of two plots, one is the plot of logarithmic of magnitude of transfer function and the other

* response given by system when ω freq. ω is changed over certain range

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is plot of phase angle. Both plots are drawn on semi-log graph using logarithmic scale.

Bode plots are most widely used for displaying the frequency response of information. So in horizontal or x-axis, it consists of frequency axis and in vertical or y-axis, it consists of wide range of gain.

To plot the magnitude in dB and phase angle in degree against $\log_{10} \omega$, logarithmic scale is used. This is available on semi log paper. In such paper, the x-axis is divided into a logarithmic scale which is non-linear while y-axis is divided into linear scale and hence it is called semi log paper.

The interesting part about x-axis is that the distance between 1 and 2 is greater than the distance between 2 and 3 and so on. Similarly, the distance between 10 and 100 & 100 and 1000 is equal. This distance is called one decade.

If $G(s) = \frac{k(a+jb)}{(a_1+jb_1)}$, then $\omega = \frac{a}{b}$ } corner
 $\omega_1 = \frac{a_1}{b_1}$ } frequency

$$\therefore \phi(\omega) = \tan^{-1}\left(\frac{b_1}{a_1}\right) - \tan^{-1}\left(\frac{b}{a}\right)$$

Types of System	Initial slope	Intersection with 0dB axis
0	0dB/decade	st line to 0dB axis
1	-20dB/decade	$\omega = k^{1/N} = k$
2	-40 dB/decade	$\omega = k^{1/2}$
3	-60 dB/decade	$\omega = k^{1/3}$

Q.1. Sketch the bode plot for $G(s) = \frac{100(1+s)}{1+0.1s}$

sol/ We have, $G(s) = \frac{100(1+s)}{1+0.1s}$

Replacing s by $j\omega$, $G(j\omega) = \frac{100(1+j\omega)}{1+0.1j\omega}$

For magnitude plot, the required frequency are:

$$\omega_1 = 1 \text{ rad/s}$$

$$\omega_2 = \frac{1}{0.1} = 10 \text{ rad/s}$$

Table for
Magnitude plot:

Factor	Corner frequency	Asymptotic log character	Resultant slope
1. 100	-	$20 \log(100) = 40 \text{ dB/decade}$ It is a straight line with magnitude 40dB/dec. and originate ^{slope} 0dB/dec	0dB/decade
2. $1+j\omega$	1rad/s	It is a st. line with ^{slope} magnitude +20dB/decade and originates from 1rad/s.	20dB/decade
3. $\frac{1}{1+j0.1\omega}$	10rad/s	It is a st. line with ^{slope} magnitude -20 dB/decade and originates from 10 rad/s.	0dB/decade

For phase plot:

$$\phi(\omega) = \tan^{-1}(\omega) - \tan^{-1}(0.1\omega)$$

ω	$\phi(\omega)$
0.1	5.13°
0.2	10.16°
0.5	23.7°
1	39.28°
2	52.125°
5	52.125°
10	39.28°
20	23.70°
30	16.5°
100	5.13°

2. Sketch the bode plot for $G(s) = \frac{10(1+0.1s)(1+0.05s)}{(1+0.01s)(1+0.001s)}$

∴ We have, $G(s) = \frac{10(1+0.1s)(1+0.05s)}{(1+0.01s)(1+0.001s)}$

Replacing s by $j\omega$, $G(j\omega) = \frac{10(1+0.1j\omega)(1+0.05j\omega)}{(1+0.01j\omega)(1+0.001j\omega)}$

For magnitude plot, the corner frequency are:

$$\omega_1 = \frac{1}{0.1} = 10 \text{ rad/s}$$

$$\omega_2 = \frac{1}{0.05} = 20 \text{ rad/s}$$

$$\omega_3 = \frac{1}{0.01} = 100 \text{ rad/s}$$

$$\omega_4 = \frac{1}{0.001} = 1000 \text{ rad/s}$$

Table for magnitude plot: $20 \log(10) = 20 \text{ dB/decade}$

Factor	Corner frequency	Asymptotic log character	Resultant slope
1. $10 \pm 1s$	10 rad/s	It is a st. line with magnitude 20 dB/decade and slope 0 dB/decade .	0 dB/decade
2. $1 + 0.1s$	10 rad/s	It is a st. line with slope magnitude and originates from 10 rad/s .	20 dB/decade
3. $1 + 0.05s$	20 rad/s	It is a st. line with slope $+20 \text{ dB/decade}$ & originates from 20 rad/s .	40 dB/decade
4. $\frac{1}{1 + 0.01s}$	100 rad/s	It is a st. line with slope -20 dB/decade & originates from 100 rad/s .	20 dB/decade
5. $\frac{1}{1 + 0.001j\omega}$	1000 rad/s	It is a st. line with slope -20 dB/decade & originates from 1000 rad/s .	0 dB/decade

For phase plot:

$$\phi(\omega) = \tan^{-1}(0.1\omega) + \tan^{-1}(0.05\omega) - \tan^{-1}(0.01\omega) - \tan^{-1}(0.001\omega)$$

ω	$\phi(\omega)$	ω	$\phi(\omega)$
1	7.94	200	96.68
2	15.76	500	71.30
5	37.45	1000	48.99
10	65.28		
20	95.97		
50	117.46		
100	112.26		

3. Sketch the bode plot for $H(s) = \frac{(1+0.2s)(1+0.025s)}{s^3(1+0.001s)(1+0.005s)}$

∴ We have,

$$H(s) = \frac{(1+0.2s)(1+0.025s)}{s^3(1+0.001s)(1+0.005s)}$$

Replacing s by $j\omega$, $H(j\omega) = \frac{(1+0.2j\omega)(1+0.025j\omega)}{(j\omega)^3(1+0.001j\omega)(1+0.005j\omega)}$

For magnitude plot, the corner frequency are:

Factor $\omega_1 = \frac{1}{0.2} = 5 \text{ rad/s}$

∴ $\omega_2 = \frac{1}{0.025} = 40 \text{ rad/s}$

$\omega_3 = \frac{1}{0.001} = 1000 \text{ rad/s}$

$\omega_4 = \frac{1}{0.005} = 200 \text{ rad/s}$

Table for magnitude plot:

$$20 \log(1) = 0$$

Factor	Corner frequency	Asymptotic log character	Res. slope
1. $\frac{1}{(j\omega)^3}$	0 rad/s	It is a straight line with slope -60 dB/decade & mag. 0 dB/decade	0 dB/decade -60 dB/dec.
2. $1+0.2j\omega$	5 rad/s	It is a st. line with slope +20 dB/dec. & originates from 5 rad/s.	-40 dB/dec.
3. $1+0.025j\omega$	40 rad/s	It is a st. line with slope +20 dB/decade and originates from 40 rad/s.	-20 dB/dec.
4. $\frac{1}{1+0.001j\omega}$	200 rad/s	It is a st line with slope -20 dB/decade & originates from 200 rad/s.	-40 dB/dec.
5. $\frac{1}{1+0.005j\omega}$	1000 rad/s	It is a st. line with slope -20 dB/dec. & originates from 1000 rad/s.	-60 dB/dec.

Here,

$$\phi(\omega) = \tan^{-1}(0.2\omega) + \tan^{-1}(0.025\omega) - \tan^{-1}(\omega) - \tan^{-1}(0.001\omega) - \tan^{-1}(0.005\omega)$$

$$\therefore \phi(\omega) = \tan^{-1}(0.2\omega) + \tan^{-1}(0.025\omega) - 90 - \tan^{-1}(0.001\omega) - \tan^{-1}(0.005\omega)$$

For phase angle: $\phi(\omega) = \tan^{-1}(0.2\omega)$

ω	$\phi(\omega)$
1	-77.6°
2	-66.02°
5	-39.59°
10	-15.96°
20	5.67°
50	28.73°
100	33.06°
400	-1.66°
800	-27.84°
1000	-36.26°

Q.4. Construct a bode plot for the given function

$$G(s) = \frac{100(s+10)}{s(s+2)(s+5)}$$

Here, $G(s) = \frac{100(s+10)}{s(s+2)(s+5)}$

Reducing the given function to standard form,

$$G(s) = \frac{100 \times 10 (0.1s+1)}{10s (0.5s+1) (0.2s+1)} = \frac{100 (0.1s+1)}{s (0.5s+1) (0.2s+1)}$$

For corner frequency:

$$\omega_1 = \frac{1}{0.1} = 10 \text{ rad/s}$$

$$\omega_2 = \frac{1}{0.5} = 2 \text{ rad/s}$$

$$\omega_3 = \frac{1}{0.2} = 5 \text{ rad/s}$$

Replacing $s \rightarrow j\omega$,

$$G(j\omega) = \frac{100 (0.1j\omega + 1)}{j\omega (0.5j\omega + 1) (0.2j\omega + 1)}$$

Table for magnitude plot:

Factor	Corner frequency	Asymptotic log character	Resultant slope
1. $\frac{100}{j\omega}$	0 -	$20 \log(100) = 40$ It is a st. line with magnitude 40 dB/dec. and slope -20 dB/dec.	-20 dB/dec.
2. $\frac{1}{(0.5j\omega + 1)}$	2 rad/s	It is a st. line with slope -20 dB/dec and originates from 2 rad/s.	-40 dB/dec.
3. $\frac{1}{(0.2j\omega + 1)}$	5 rad/s	It is a st. line with slope -20 dB/dec and originates from 5 rad/s	-60 dB/dec.
4. $0.1j\omega + 1$	10 rad/s	It is a st. line with slope +20 dB/dec. and originates from 10 rad/s.	-40 dB/dec.

For phase plot:

$$\phi(\omega) = \tan^{-1}(0.1\omega) - 90^\circ - \tan^{-1}(0.5) - \tan^{-1}(0.2)$$

$$\therefore \phi(\omega) = \tan^{-1} \phi(\omega)$$

$$0.1$$

$$5. \quad G(s) = \frac{20(s+2)}{s(s^2+4s+16)}$$

$$\text{Here, } G(s) = \frac{20(s+2)}{s(s^2+4s+16)}$$

$$= \frac{20 \times 2(0.5s+1)}{s \times 16 \left(\frac{s^2}{16} + \frac{s}{4} + 1 \right)}$$

$$\therefore G(s) = \frac{2.5(0.5s+1)}{s \left(\frac{s^2}{16} + \frac{s}{4} + 1 \right)}$$

Comparing $\frac{s^2}{16} + \frac{s}{4} + 1 = 0$ with $\frac{s^2}{\omega_n^2} + 2\zeta \frac{s}{\omega_n} + 1 = 0$, we get,

$$\omega_n^2 = 4^2 \quad \therefore \omega_n = 4$$

$$\text{And, } \frac{2\zeta}{\omega_n} = \frac{1}{4} \quad \therefore \zeta = 0.5$$

For corner frequency,

$$\omega_1 = \frac{1}{0.5} = 2 \text{ rad/s}$$

$$\omega_2 = 4 \text{ rad/s}$$

Replacing $s \rightarrow j\omega$, $G(j\omega) = \frac{2.5(0.5j\omega+1)}{j\omega \left(\frac{j^2\omega^2}{16} + \frac{j\omega}{4} + 1 \right)}$

Table for magnitude plot:

Factor	Corner frequency	Asymptotic log character	Resultant slope
1. $\frac{2.5}{j\omega}$	—	$20 \log(2.5) = 7.95$ It is a st. line with magnitude 7.95 dB & slope -20 dB/dec.	-20 dB/dec.
2. $0.5j\omega+1$	2 rad/s	It is a st. line with slope +20 dB/dec & originates from 20 rad/s.	0 dB/dec.
3. $\frac{1}{\left(\frac{j^2\omega^2}{16} + \frac{j\omega}{4} + 1 \right)}$	4 rad/s	It is a st. line with slope -40 dB/dec & originates from 4 rad/s.	-40 dB/dec

We have, $s^2 + 4s + 16 = 0$
 $\Rightarrow j^2 \omega^2 + 4j\omega + 16 = 0$
 $\Rightarrow -\omega^2 + 4j\omega + 16 = 0$
 $\Rightarrow (-\omega^2 + 16) + 4j\omega = 0$

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For phase plot:

$$\phi(\omega) = \tan^{-1}(0.5\omega) - 90 - \tan^{-1}\left(\frac{4\omega}{16 - \omega^2}\right)$$

ϕ	ω	$\phi(\omega)$
	0.1	

6. $N(s) = \frac{20(s+1)}{s(s+5)(s^2+4s+16)}$

Sol: Here, $N(s) = \frac{20(s+1)}{s(s+5)(s^2+4s+16)}$

$$= \frac{20(s+1)}{5 \times 16(s(0.25s+1)\left(\frac{s^2}{16} + \frac{s}{4} + 1\right))}$$

$$\therefore N(s) = \frac{0.25(s+1)}{s(0.25s+1)\left(\frac{s^2}{16} + \frac{s}{4} + 1\right)}$$

Comparing $\frac{s^2}{16} + \frac{s}{4} + 1 = 0$ with $\frac{s^2}{\omega_n^2} + \frac{2\zeta_s}{\omega_n}s + 1 = 0$,
 we get, $\omega_n^2 = 4^2 \therefore \omega_n = 4$

And, $\frac{2\zeta_s}{\omega_n} = \frac{1}{4} \therefore \zeta_s = 0.5$

For corner frequencies:

$$\omega_1 = 1 \text{ rad/s}$$

$$\omega_3 = 4 \text{ rad/s}$$

$$\omega_2 = \frac{1}{0.2} = 5 \text{ rad/s}$$

Replacing $s \rightarrow j\omega$, $N(j\omega) = \frac{0.25(j\omega + 1)}{j\omega(0.2j\omega + 1)\left(\frac{j^2\omega^2}{16} + \frac{j\omega}{4} + 1\right)}$

Table for magnitude plot:

Factor	Corner frequency	Asymptotic log character	Resultant slope
1. $\frac{0.25}{j\omega}$	—	$20 \log(0.25) = -12.04 \text{ dB}$ It is a st. line with slope magnitude -12.04 dB with slope -20 dB/dec .	-20 dB/dec
2. $j\omega + 1$	1 rad/s	It is a st. line with magnitude slope $+20 \text{ dB/dec}$ and originates from 1 rad/s .	0 dB/dec
3. $\frac{1}{0.2j\omega + 1}$	5 rad/s	It is a st. line with slope -20 dB/dec & originates from 5 rad/s .	-20 dB/dec
4. $\frac{1}{\left(\frac{j^2\omega^2}{16} + \frac{j\omega}{4} + 1\right)}$	4 rad/s	It is a st. line with slope -40 dB/dec and originates from 4 rad/s .	-40 dB/dec

We have, $s^2 + 4s + 16 = 0$

$\therefore (\omega^2 - 16) + (4j\omega) = 0$

$\therefore \phi(\omega) = \tan^{-1}(\omega) - 90 - \tan^{-1}(0.2\omega) - \tan^{-1}\left(\frac{4\omega}{16 - \omega^2}\right)$

	ω	$\phi(\omega)$	ω	$\phi(\omega)$
1.	0.1		8.	20
2.	0.2		9.	30
3.	0.5		10.	100
4.	1			
5.	2			
6.	5			
7.	10			

7. $G(s) = \frac{300(s+4)}{5(2s+1)(s+10)(3s^2+11s+12)}$

Here, $G(s) = \frac{300(s+4)}{5(2s+1)(3s^2+11s+12)(s+10)}$

Reducing it into standard form,

$$G(s) = \frac{300 \times 4 (0.25s+1)}{5 \times 10 \times \frac{12}{3} (2s+1) \left(\frac{s^2}{4} + \frac{11s}{12} + 1 \right) (0.1s+1)}$$

Comparing $\frac{s^2}{4} + \frac{11s}{12} + 1 = 0$ with $\frac{s^2}{\omega_n^2} + \frac{2\zeta_s}{\omega_n}s + 1 = 0$,

$$\omega_n = 2 \text{ rad/s.}$$

$$\text{And, } \frac{2\zeta_s}{\omega_n} = \frac{11}{4} \Rightarrow \frac{2\zeta_s}{2} = \frac{11}{12} \therefore \zeta_s = \frac{11}{12}$$

For corner frequencies,

$$\omega_1 = \frac{1}{0.25} = 4 \text{ rad/s}$$

$$\omega_2 = \frac{1}{2} = 0.5 \text{ rad/s}$$

$$\omega_3 = 2 \text{ rad/s}$$

$$\omega_4 = \frac{1}{0.1} = 10 \text{ rad/s.}$$

Replacing s by $j\omega$, $G(j\omega) = \frac{10(0.25j\omega+1)}{j\omega(2j\omega+1)\left(\frac{j^2\omega^2}{4} + \frac{11j\omega}{12} + 1\right)(0.1j\omega+1)}$

Table for magnitude plot:

Factor	Corner freq.	Asymptotic log character	Resultant slope
1. $\frac{10}{j\omega}$	—	$20 \log(10) = 20 \text{ dB}$ It is a st. line with magnitude 20 dB & slope -20 dB/dec.	-20 dB/dec.

2. $\frac{1}{2s+1}$ 0.5 rad/s It is a st. line with -20 dB/dec. magnitude slope -20 dB/dec and originates from 0.5 rad/s.

4. $0.25j\omega+1$ 4 rad/s It is a st. line with -20 dB/dec. slope +20 dB/dec. and originates from 4 rad/s

3. $\frac{1}{(\frac{j^2\omega^2}{4} + \frac{11j\omega}{12} + 1)}$ 2 rad/s It is a st. line with -60 dB/dec. slope -40 dB/dec and originates from 2 rad/s.

5. $\frac{1}{(0.1j\omega+1)}$ 10 rad/s It is a st. line with -80 dB/dec. slope -20 dB/dec. and originates from 10 rad/s

$$\text{We have, } (3s^2 + 11s + 12) = 0 \Rightarrow (12 - 3\omega^2) + 11j\omega = 0$$

For phase plot:

$$\phi(\omega) = \tan^{-1}(0.25j\omega) - 90 - \tan^{-1}(2j\omega) - \tan^{-1}(0.1j\omega) - \tan^{-1}\left(\frac{11\omega}{12 - 3\omega^2}\right)$$

$\therefore \omega$ $\phi(\omega)$

0.1

0.2

0.5

1

2

5

10

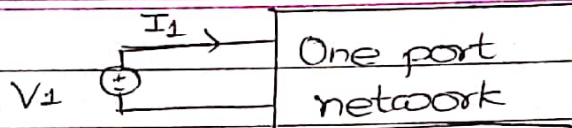


fig- Block diagram of one port network

The basic idea of one port network is as shown above. The one port network is a blackbox with a single pair of i/p terminals referred to as port. A force or voltage is applied at the terminal and velocity or current flows in any direction. Network theory is normally described in terms of circuit elements in which voltage is applied at terminals and the current flows.

A single port network contains a source that excites it. The ratio of source voltage to the current drawn is called driving point impedance. i.e. $Z(s) = \frac{V(s)}{I(s)}$.

Similarly, driving point admittance, $Y(s) = \frac{I(s)}{V(s)}$

Synthesis of one port network: The ^{elements of} synthesis of one port network with are RL, LC or RC, we will proceed and is obtained by partial fraction expansion and continued fraction expansion. of given function.

Methods =

1. Foster form

2. Cauer form

Cauer form - The Cauer I and Cauer II form are obtained by continued fraction expansion of function by arranging both numerator and denominator in descending order and ascending order respectively.

Q1. Synthesize the function $Z(s) = \frac{2s^5 + 12s^3 + 16s}{s^4 + 4s^2 + 3}$ using Cauer form.

For Cauer's first form, $Z(s)$ must be in descending order.

$$Z(s) = \frac{2s^5 + 12s^3 + 16s}{s^4 + 4s^2 + 3}$$

$$\begin{array}{r} s^4 + 4s^2 + 3 \overline{) 2s^5 + 12s^3 + 16s} \\ \underline{2s^5 + 8s^3 + 6s} \end{array}$$

$$\begin{array}{r} 2s^3 + 10s \overline{) s^4 + 4s^2 + 3} \\ \underline{s^4 + 10s^2} \end{array}$$

$$\begin{array}{r} 3s^2 + 3 \overline{) 4s^3 + 10s} \\ \underline{4s^3 + 8s} \end{array}$$

$$\begin{array}{r} 2s \overline{) 3s^2 + 3} \\ \underline{3s^2 + 3} \end{array}$$

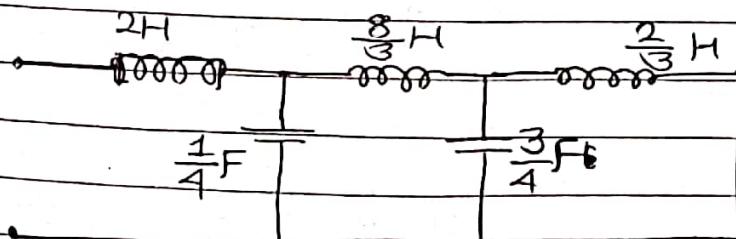
Here, $Z_1(s) = 2s \Rightarrow Ls = 2s \therefore L = 2H$

$Z_2(s) = \frac{s}{4} \Rightarrow Cs = \frac{s}{4} \therefore C = \frac{1}{4}F$

$Z_3(s) = \frac{8s}{3} \Rightarrow Ls = \frac{8s}{3} \therefore L = \frac{8}{3}H$

$Z_4(s) = \frac{3s}{4} \Rightarrow Cs = \frac{3s}{4} \therefore C = \frac{3}{4}F$

$Z_5(s) = \frac{2}{3} \Rightarrow Ls = \frac{2}{3} \therefore L = \frac{2}{3}H$



Cauer first form of $Z(s)$

For Cauchy's second form, $Z(s)$ must be in ascending order.

$$\therefore Z(s) = \frac{16s + 12s^3 + 2s^5}{3 + 4s^2 + s^4}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) 16s + 12s^3 + 2s^5} \quad \left(\frac{16s}{3} - \frac{28s^3}{3} \right) \\ \underline{16s + 64s^3 + 16s^5} \\ -28s^3 - 10s^5 \end{array}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) -28s^3 - 10s^5} \quad \left(\frac{-28s^3}{3} - \frac{10s^5}{3} \right) \\ \underline{-28s^3 - 15s^5} \\ 5s^5 \end{array}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) 5s^5} \quad \left(\frac{5s^5}{3} - \frac{20s^7}{3} \right) \\ \underline{5s^5 - 20s^7} \\ 20s^7 \end{array}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) 20s^7} \quad \left(\frac{20s^7}{3} - \frac{80s^9}{3} \right) \\ \underline{20s^7 - 80s^9} \\ 80s^9 \end{array}$$

Here, $Z_1(s) = \frac{16s}{3} \therefore L = \frac{16}{3} H$

$$Z_2(s) = \frac{-28s^3}{3}$$

$$Y(s) = \frac{-28s^3}{3 + 4s^2 + s^4}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) -28s^3} \quad \left(\frac{-28s^3}{3} - \frac{10s^5}{3} \right) \\ \underline{-28s^3 - 10s^5} \\ 10s^5 \end{array}$$

Here, $Y_1(s) = \frac{3}{16s} \therefore L = \frac{16}{3} H$

$$Z_2(s) = \frac{54s}{7} \therefore L = \frac{54}{7} H$$

$$Y_3(s) = \frac{7}{176s}$$

$$\therefore L = \frac{176}{49} H$$

$$Z_4(s) =$$

$$\begin{array}{r} \frac{7}{4}s^2 + \frac{5}{8}s^4 \overline{) 16s + 12s^3 + 2s^5} \quad \left(\frac{16s}{7} - \frac{4s^3}{7} \right) \\ \underline{16s + 40s^3} \\ -28s^3 - 10s^5 \end{array}$$

$$\begin{array}{r} \frac{7}{4}s^2 + \frac{5}{8}s^4 \overline{) -28s^3 - 10s^5} \quad \left(\frac{-28s^3}{7} - \frac{10s^5}{7} \right) \\ \underline{-28s^3 - 15s^5} \\ 5s^5 \end{array}$$

$$\begin{array}{r} \frac{3}{44}s^4 \overline{) 44s^3 + 2s^5} \quad \left(\frac{44s^3}{7} - \frac{44s^5}{7} \right) \\ \underline{44s^3 - 44s^5} \\ 44s^5 \end{array}$$

$$\begin{array}{r} 2s^5 \overline{) 3s^4} \quad \left(\frac{3s^4}{44} - \frac{3s^6}{44} \right) \\ \underline{3s^4 - 3s^6} \\ 3s^6 \end{array}$$

Q. The given function is $Z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)}$

$$\Rightarrow Z(s) = \frac{s^2 + 5s + 4}{s(s^2 + 7s + 10)}$$

$$\therefore Z(s) = \frac{s^2 + 5s + 4}{s^3 + 7s^2 + 10s}$$

Cauer-I form:

$$\begin{array}{r} s^3 + 7s^2 + 10s \overline{) s^2 + 5s + 4} \left(\frac{1}{s} \right. \\ \underline{s^2 + 7s + 10} \\ -2s - 6 \end{array}$$

Since the remainder is negative, Cauer-I form cannot be obtained from the given function $Z(s)$.